39

Relativity

CHAPTER OUTLINE

39.1 The Principle of Galilean Relativity

39.2 The Michelson-Morley Experiment

39.3 Einstein's Principle of Relativity

39.4 Consequences of the Special Theory of Relativity

39.5 The Lorentz Transformation Equations

39.6 The Lorentz Velocity Transformation Equations

39.7 Relativistic Linear Momentum

39.8 Relativistic Energy

39.9 The General Theory of Relativity

* An asterisk indicates a question or problem new to this edition.

ANSWERS TO OBJECTIVE QUESTIONS

OQ39.1 **(i)** Answer (a). **(ii)** Answer (c). **(iii)** Answer (d). There is no upper limit on the momentum or energy of an electron. As the speed of the electron approaches c, the factor γ tends to infinity, so both the kinetic energy, $K = (\gamma - 1)mc^2$, and momentum, $p = \gamma mv$, tend to infinity.

OQ39.2 Answer (d). The relativistic time dilation effect is symmetric between the observers.

OQ39.3 Answers (b) and (c). According to the second postulate of special relativity (the constancy of the speed of light), both observers will measure the light speed to be c.

OQ39.4 Answer (c). An oblate spheroid. The dimension in the direction of motion would be contracted but the dimension perpendicular to the motion would be unaltered.

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

OQ39.5 Answer (e). The astronaut is moving with constant velocity and is therefore in an inertial reference frame. According to the principle of relativity, all the laws of physics are the same in her reference frame as in any other inertial reference frame. Thus, she should experience no effects due to her motion through space.

OQ39.6 Answer (b). The dimension parallel to the direction of motion is reduced by the factor γ and the other dimensions are unchanged.

OQ39.7 **(i)** Answer (c). The Earth observer measures the clock in orbit to run slower.

(ii) Answer (b). They are not synchronized. They both tick at the same rate after return, but a time difference has developed between the two clocks.

OQ39.8 Answer (a) > (c) > (b). The relativistic momentum of a particle is $p = \sqrt{E^2 - E_R^2}/c$, where E is the total energy of the particle, and $E_R = mc^2$ is its rest energy ($E_R = 0$ for the photon). In this problem, each of the particles has the same total energy E. Thus, the particle with the smallest rest energy (photon < electron < proton) has the greatest momentum.

OQ39.9 Answers (d) and (e). The textbook refers to the postulate summarized in choice (d) as the principle of relativity, and to the postulate in choice (e) as the constancy of the speed of light.

OQ39.10 Answer (b). By the postulate of the constancy of the speed of light, light from any source travels in vacuum at speed c.

ANSWERS TO CONCEPTUAL QUESTIONS

CQ39.1 The star and the planet orbit about their common center of mass, thus the star moves in an elliptical path. Just like the light from a star in a binary star system, the spectrum of light from the star would undergo a cyclic series of Doppler shifts depending on the star's speed and direction of motion relative to the observer. The repetition rate of the Doppler shift pattern is the period of the orbit. Information about the orbit size can be calculated from the size of the Doppler shifts.

CQ39.2 Suppose a railroad train is moving past you. One way to measure its length is this: You mark the tracks at the cowcatcher forming the front of the moving engine at 9:00:00 AM, while your assistant marks the tracks at the back of the caboose at the same time. Then you find the distance between the marks on the tracks with a tape measure.

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

You and your assistant must make the marks simultaneously in your frame of reference, for otherwise the motion of the train would make its length different from the distance between marks.

CQ39.3 (a) Yours does. From your frame of reference, the clocks on the train run slow, so the symphony takes a longer time interval to play on the train.

(b) The observer's on the train does. From the train's frame of reference, your clocks run slow, so the symphony takes a longer time interval to play for you.

(c) Each observer measures his symphony as finishing first.

CQ39.4 Get a *Mr. Tompkins* book by George Gamow for a wonderful fictional exploration of this question. Because of time dilation, your trip to work would be short, so your coffee would not have time to become cold, and you could leave home later. Driving home in a hurry, you push on the gas pedal not to increase your speed by very much, but rather to make the blocks get shorter. Big Doppler shifts in wave frequencies make red lights look green as you approach them, alter greatly the frequencies of car horns, and make it very difficult to tune a radio to a station. High-speed transportation is very expensive because a small change in speed requires a large change in kinetic energy, resulting in huge fuel use. Crashes would be disastrous because a speeding car has a great amount of kinetic energy, so a collision would generate great damage. There is a five-day delay in transmission when you watch the Olympics in Australia on live television. It takes ninety-five years for sunlight to reach Earth.

CQ39.5 Acceleration is indicated by a curved line. This can be seen in the middle of Speedo's world-line in Figure 39.11, where he turns around and begins his trip home.

CQ39.6 (a) Any physical theory must agree with experimental measurements within some domain. Newtonian mechanics agrees with experiment for objects moving slowly compared to the speed of light. Relativistic mechanics agrees with experiment for objects moving at relativistic speeds.

(b) It is well established that Newtonian mechanics applies to objects moving at speeds a lot less than light, but Newtonian mechanics fails at relativistic speeds. If relativistic mechanics is to be the better theory, it must apply to all physically possible speeds. Relativistic mechanics at nonrelativistic speeds must reduce to Newtonian mechanics, and it does.

CQ39.7 No. The principle of relativity implies that nothing can travel faster than the speed of light in a *vacuum*, which is 300 Mm/s. The electron would emit light in a conical shock wave of Cerenkov radiation.

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

CQ39.8 According to $\vec{\mathbf{p}} = \gamma m \vec{\mathbf{u}}$, doubling the speed u will make the momentum of an object increase by the factor $2\left[\dfrac{c^2 - u^2}{c^2 - 4u^2}\right]^{1/2}$.

CQ39.9 As the object approaches the speed of light, its kinetic energy grows without limit. It would take an infinite investment of work to accelerate the object to the speed of light.

CQ39.10 A microwave pulse is reflected from a moving object. The waves that are reflected back are Doppler shifted in frequency according to the speed of the target. The receiver in the radar gun detects the reflected wave and compares its frequency to that of the emitted pulse. Using the frequency shift, the speed can be calculated to high precision. Be forewarned: this technique works if you are either traveling toward or away from your local law enforcement agent!

CQ39.11 Running "at a speed near that of light" means some other observer measures you to be running near the speed of light. To you, you are at rest in your own inertial frame. You would see the same thing that you see when looking at a mirror when at rest. The theory of relativity tells us that all experiments will give the same results in all inertial frames of reference.

CQ39.12 **(i)** Solving for the image location q in terms of the object location p and the focal length f gives

$$q = \frac{pf}{p - f}$$

We note that when $p = f$, the image is formed at infinity. Let us, for example, take an object initially a distance $p_i = 2f$ from the mirror. Its speed, in approaching f in a finite amount of time is

$$v = \frac{p - f}{\Delta t} = \frac{2f - f}{\Delta t} = \frac{f}{\Delta t}$$

At the same time, the location of the image moves from $q_i = (2f)f\,/\,(2f - f) = 2f$ to $q_f = \infty$, i.e., covering an infinite distance in a finite amount of time. The speed of the image thus exceeds the speed of light c.

(ii) For simplicity, we assume that the distant screen is curved with a radius of curvature R. The linear speed of the spot on the screen is then given by $v = \omega R$, where ω is the angular speed of rotation of the laser pointer. With sufficiently large ω and R, the speed of the spot moving on the screen can exceed c.

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

(iii) Neither of these examples violates the principle of relativity. In the first case, the image transtions from being real to being virtual when $p = f$. In the second case, we have the intersection of a light beam with a screen. A point of tranition or intersection is not made of matter so it has no mass, and hence no energy. A bug momentarily at the intersection point could squeak or reflect light. A second bug would have to wait for sound or light to travel across the distance between the first bug and himself, to get the message; neither of these actions would result in communication reaching the second bug sooner than the intersection point reaches him.

CQ39.13 Special relativity describes the relationship between physical quantities and laws in inertial reference frames: that is, reference frames that are not accelerating. General relativity describes the relationship between physical quantities and laws in all reference frames.

CQ39.14 Because of gravitational time dilation, the downstairs clock runs more slowly because it is closer to the Earth and hence in a stronger gravitational field than the upstairs clock.

SOLUTIONS TO END-OF-CHAPTER PROBLEMS

Section 39.1 The Principle of Galilean Relativity

P39.1 By Equation 4.20, $\vec{\mathbf{u}}_{PA} = \vec{\mathbf{u}}_{PB} + \vec{\mathbf{v}}_{BA}$, with motion in one dimension,

$$u_{baseball,\,\text{ground}} = u_{baseball,\,\text{truck}} + \vec{\mathbf{v}}_{\text{truck, ground}}$$

$$u_{baseball,\,\text{ground}} = -20.0\ \text{m/s} + 10.0\ \text{m/s} = -10.0\ \text{m/s}$$

In other words, $\boxed{10.0\ \text{m/s toward the left in Figure P39.1}}$.

P39.2 In the laboratory frame of reference, Newton's second law is valid: $\vec{\mathbf{F}} = m\vec{\mathbf{a}}$. Laboratory observer 1 watches some object accelerate under applied forces. Call the instantaneous velocity of the object $\vec{\mathbf{v}}_1 = \vec{\mathbf{v}}_{O1}$ (the velocity of object O relative to observer 1 in laboratory frame) and its acceleration $\dfrac{d\vec{\mathbf{v}}_1}{dt} = \vec{\mathbf{a}}_1$. A second observer has instantaneous velocity $\vec{\mathbf{v}}_{21}$ relative to the first. In general, the velocity of the object in the frame of the second observer is

$$\vec{\mathbf{v}}_2 = \vec{\mathbf{v}}_{O2} = \vec{\mathbf{v}}_{O1} + \vec{\mathbf{v}}_{12} = \vec{\mathbf{v}}_1 - \vec{\mathbf{v}}_{21}$$

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

(a) If the relative instantaneous velocity $\vec{\mathbf{v}}_{21}$ of the second observer is *constant*, the second observer measures the acceleration

$$\vec{\mathbf{a}}_2 = \frac{d\vec{\mathbf{v}}_2}{dt} = \frac{d\vec{\mathbf{v}}_1}{dt} = \vec{\mathbf{a}}_1$$

This is the same as that measured by the first observer. In this nonrelativistic case, they measure the same forces and masses as well. Thus, the second observer also confirms that $\vec{\mathbf{F}} = m\vec{\mathbf{a}}$.

(b) If the second observer's frame is accelerating, then the instantaneous relative velocity $\vec{\mathbf{v}}_{21}$ is *not constant*. The second observer measures an acceleration of

$$\vec{\mathbf{a}}_2 = \frac{d\vec{\mathbf{v}}_2}{dt} = \frac{d\left(\vec{\mathbf{v}}_1 - \vec{\mathbf{v}}_{21}\right)}{dt} = \vec{\mathbf{a}}_1 - \frac{d\left(\vec{\mathbf{v}}_{21}\right)}{dt} = \vec{\mathbf{a}}_1 - \vec{\mathbf{a}}',$$

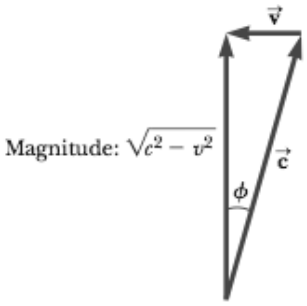
where $\dfrac{d\left(\vec{\mathbf{v}}_{21}\right)}{dt} = \vec{\mathbf{a}}'$

The observer in the accelerating frame measures the acceleration of the mass as being $\vec{\mathbf{a}}_2 = \vec{\mathbf{a}}_1 - \vec{\mathbf{a}}'$. If Newton's second law held for the accelerating frame, that observer would expect to find valid the relation $\vec{\mathbf{F}}_2 = m\vec{\mathbf{a}}_2$, or $\vec{\mathbf{F}}_1 = m\vec{\mathbf{a}}_2$ (since $\vec{\mathbf{F}}_1 = \vec{\mathbf{F}}_2$ and the mass is unchanged in each). But, instead, the accelerating frame observer finds that $\vec{\mathbf{F}}_2 = m\vec{\mathbf{a}}_2 - m\vec{\mathbf{a}}'$, which is *not* Newton's second law.

P39.3 From the triangle in ANS. FIG. P39.3,

$$\phi = \sin^{-1}\left(\frac{v}{c}\right) = \sin^{-1}\left(\frac{29.8\times10^3 \text{ m/s}}{2.998\times10^8 \text{ m/s}}\right)$$

$$= \boxed{5.70\times10^{-3} \text{ degrees} = 9.94\times10^{-5} \text{ rad}}$$



ANS. FIG. P39.3

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

P39.4 In the rest frame,

$$p_i = m_1 v_{1i} + m_2 v_{2i} = (2\,000\ \text{kg})(20.0\ \text{m/s}) + (1\,500\ \text{kg})(0\ \text{m/s})$$
$$= 4.00 \times 10^4\ \text{kg} \cdot \text{m/s}$$
$$p_f = (m_1 + m_2) v_f = (2\,000\ \text{kg} + 1\,500\ \text{kg}) v_f$$

Since $p_i = p_f$,

$$v_f = \frac{p_i}{m_1 + m_2} = \frac{4.00 \times 10^4\ \text{kg} \cdot \text{m/s}}{2\,000\ \text{kg} + 1\,500\ \text{kg}} = 11.429\ \text{m/s}$$

In the moving frame, these velocities are all reduced by +10.0 m/s.

$$v'_{1i} = v_{1i} - v' = 20.0\ \text{m/s} - (+10.0\ \text{m/s}) = 10.0\ \text{m/s}$$
$$v'_{2i} = v_{2i} - v' = 0\ \text{m/s} - (+10.0\ \text{m/s}) = -10.0\ \text{m/s}$$
$$v'_f = 11.429\ \text{m/s} - (+10.0\ \text{m/s}) = 1.429\ \text{m/s}$$

Our initial momentum is then

$$p'_i = m_1 v'_{1i} + m_2 v'_{2i}$$
$$= (2\,000\ \text{kg})(10.0\ \text{m/s}) + (1\,500\ \text{kg})(-10.0\ \text{m/s})$$
$$= 5\,000\ \text{kg} \cdot \text{m/s}$$

and our final momentum has the same value:

$$p'_f = (2\,000\ \text{kg} + 1\,500\ \text{kg}) v'_f = (3\,500\ \text{kg})(1.429\ \text{m/s})$$
$$= 5\,000\ \text{kg} \cdot \text{m/s}$$

Section 39.2 The Michelson-Morley Experiment

Section 39.3 Einstein's Principle of Relativity

Section 39.4 Consequences of the Special Theory of Relativity

P39.5 In the rest frame of the spacecraft, the Earth-star gap travels past it at speed u. The distance from Earth to the star is a proper length in the Earth's frame:

$$L = \frac{L_P}{\gamma} = L_P \sqrt{1 - \left(\frac{u}{c}\right)^2}$$

Solving for the speed of the spacecraft gives,

$$u = c\sqrt{1 - \left(\frac{L}{L_P}\right)^2} = c\sqrt{1 - \left(\frac{2.00\ \text{ly}}{5.00\ \text{ly}}\right)^2} = \boxed{0.917c}$$

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

P39.6 (a) The length of the meter stick measured by the observer moving at speed $v = 0.900\,c$ relative to the meter stick is

$$L = L_p/\gamma = L_p\sqrt{1-(v/c)^2} = (1.00\text{ m})\sqrt{1-(0.900)^2} = \boxed{0.436\text{ m}}$$

(b) If the observer moves relative to Earth in the direction opposite the motion of the meter stick relative to Earth, the velocity of the observer relative to the meter stick is greater than that in part (a). The measured length of the meter stick will be $\boxed{\text{less than } 0.436\text{ m}}$ under these conditions, but so small it is unobservable.

P39.7 A clock running at one-half the rate of a clock at rest takes twice the time to register the same time interval: $\Delta t = 2\Delta t_p$.

$$\Delta t = \frac{\Delta t_p}{\left[1-(v/c)^2\right]^{1/2}} \qquad \text{so} \qquad v = c\left[1-\left(\frac{\Delta t_p}{\Delta t}\right)^2\right]^{1/2}$$

For $\Delta t = 2\Delta t_p$,

$$v = c\left[1-\left(\frac{\Delta t_p}{2\Delta t_p}\right)^2\right]^{1/2} = c\left[1-\frac{1}{4}\right]^{1/2} = \boxed{0.866c}$$

P39.8 For $\dfrac{v}{c} = 0.990$, $\gamma = 7.09$.

(a) The muon's lifetime as measured in the Earth's rest frame is

$$\Delta t = \frac{L_p}{v} = \frac{4.60\text{ km}}{0.990c} = \left[\frac{4.60\times10^3\text{ m}}{0.990\left(3.00\times10^8\text{ m/s}\right)}\right]$$
$$= 1.55\times10^{-5}\text{ s} = 15.5\ \mu\text{s}$$

and the lifetime measured in the muon's rest frame is

$$\Delta t_p = \frac{\Delta t}{\gamma} = \frac{1}{7.09}(15.5\ \mu\text{s}) = \boxed{2.18\ \mu\text{s}}$$

(b) In the muon's frame, the Earth is approaching the muon at speed $v = 0.990c$. During the time interval the muon exists, the Earth travels the distance

$$d = v\Delta t_p = v\frac{\Delta t}{\gamma} = v\frac{L_p}{\gamma v} = \frac{L_p}{\gamma}$$
$$= \left(4.60\times10^3\text{ m}\right)\sqrt{1-(0.990)^2} = \boxed{649\text{ m}}$$

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

P39.9 From Equation 39.9 for length contraction,

$$L = L_p\sqrt{1-\frac{v^2}{c^2}}$$

we solve for the speed v of the meterstick:

$$v = c\sqrt{1-\left(\frac{L}{L_p}\right)^2}$$

Taking $L = \frac{L_p}{2}$ where, $L_p = 1.00$ m, gives

$$v = c\sqrt{1-\left(\frac{L_p/2}{L_p}\right)^2} = c\sqrt{1-\frac{1}{4}} = \boxed{0.866c}$$

P39.10 (a) The time interval between pulses as measured by the astronaut is a proper time:

$$\Delta t_p = \left(\frac{1\text{ min}}{75.0\text{ beats}}\right)$$

The time interval between pulses as measured by the Earth observer is then:

$$\Delta t = \gamma\Delta t_p = \frac{1}{\sqrt{1-(0.500)^2}}\left(\frac{1\text{ min}}{75.0\text{ beats}}\right) = 1.54\times10^{-2}\text{ min/beat}$$

Thus, the Earth observer records a pulse rate of

$$\frac{1}{\Delta t} = \frac{1}{\gamma\Delta t_p} = \sqrt{1-(0.500)^2}\left(\frac{75.0\text{ beats}}{1\text{ min}}\right) = \boxed{65.0\,\text{beats/min}}$$

(b) From part (a), the pulse rate is

$$\frac{1}{\Delta t} = \frac{1}{\gamma\Delta t_p} = \sqrt{1-(0.990)^2}\left(\frac{75.0\text{ beats}}{1\text{ min}}\right) = \boxed{10.5\,\text{beats/min}}$$

That is, the life span of the astronaut (reckoned by the duration of the total number of his heartbeats) is much longer as measured by an Earth clock than by a clock aboard the space vehicle.

P39.11 For the light as observed, $\lambda = 650$ nm and $\lambda' = 520$ nm. From Equation 39.10,

$$f' = \frac{c}{\lambda'} = \sqrt{\frac{1+v/c}{1-v/c}}f = \sqrt{\frac{1+v/c}{1-v/c}}\frac{c}{\lambda}$$

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

Solving for the velocity,

$$\sqrt{\frac{1+v/c}{1-v/c}}=\frac{\lambda}{\lambda'} \quad \rightarrow \quad 1+\frac{v}{c}=\left(\frac{\lambda}{\lambda'}\right)^2\left(1-\frac{v}{c}\right)$$

Then,

$$\frac{v}{c}\left[1+\left(\frac{\lambda}{\lambda'}\right)^2\right]=\left(\frac{\lambda}{\lambda'}\right)^2-1$$

$$\frac{v}{c}=\frac{\left(\frac{\lambda}{\lambda'}\right)^2-1}{1+\left(\frac{\lambda}{\lambda'}\right)^2}=\frac{\left(\frac{650\text{ nm}}{520\text{ nm}}\right)^2-1}{1+\left(\frac{650\text{ nm}}{520\text{ nm}}\right)^2}=0.220$$

or $$v=\boxed{0.220c}=6.59\times10^7\text{ m/s}$$

P39.12 The spacecraft are identical, so they have the same proper length; thus, your measurements and the astronaut's measurements are reciprocal.

(a) You measure the proper length of your spacecraft to be

$$\boxed{L_p=20.0\text{ m}}$$

(b) You measure the length L of the astronaut's spacecraft to be

$$\boxed{L=19.0\text{ m}}$$

(c) From the astronaut's measurement of the length L of your spacecraft,

$$L=\frac{L_p}{\gamma}=L_p\sqrt{1-\left(\frac{u}{c}\right)^2}$$

we solve for the speed of the astronaut's spacecraft relative to yours:

$$\frac{u}{c}=\sqrt{1-\left(\frac{L}{L_p}\right)^2}=\sqrt{1-\left(\frac{19.0\text{ m}}{20.0\text{ m}}\right)^2}=0.312$$

or $$u=\boxed{0.312c}$$

P39.13 The astronaut's measured time interval is a proper time in her reference frame. Therefore, according to an observer on Earth,

$$\Delta t=\gamma\,\Delta t_p=\frac{\Delta t_p}{\sqrt{1-(v/c)^2}}=\frac{3.00\text{ s}}{\sqrt{1-(0.800)^2}}=\boxed{5.00\text{ s}}$$

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

P39.14 From the definition of γ,

$$\gamma = \frac{1}{\sqrt{1-\left(v^2/c^2\right)}} = 1.010\ 0$$

we solve for the speed:

$$v = c\sqrt{1-\left(\frac{1}{\gamma}\right)^2} = c\sqrt{1-\left(\frac{1}{1.010\ 0}\right)^2} = \boxed{0.140c}$$

P39.15 The observer measures the proper length of the tunnel, 50.0 m, but measures the train contracted to length

$$L = L_p\sqrt{1-\frac{v^2}{c^2}} = 100\text{ m}\sqrt{1-(0.950)^2} = 31.2\text{ m}$$

shorter than the tunnel by 50.0 – 31.2 = 18.8 m.

The trackside observer measures the length to be 31.2 m, so the supertrain is measured to fit in the tunnel, with 18.8 m to spare.

***P39.16** (a) The lifetime of the pi meson measured by an observer on Earth is given by

$$\Delta t = \gamma\Delta t_p = \frac{\Delta t_p}{\sqrt{1-(v/c)^2}} = \frac{2.6\times10^{-8}\text{ s}}{\sqrt{1-(0.98)^2}} = \boxed{1.3\times10^{-7}\text{ s}}$$

(b) The distance travelled before the meson decays is

$$d = v\Delta t = 0.98\left(3.0\times10^8\text{ m/s}\right)\left(1.3\times10^{-7}\text{ s}\right) = \boxed{38\text{ m}}$$

(c) In the absence of time dilation, the meson would travel a distance

$$d = v\Delta t = 0.98\left(3.0\times10^8\text{ m/s}\right)\left(2.6\times10^{-8}\text{ s}\right) = \boxed{7.6\text{ m}}$$

***P39.17** (a) The 0.800c and the 20.0 ly are measured in the Earth frame, so in this frame,

$$\Delta t = \frac{x}{v} = \frac{20.0\text{ ly}}{0.800c} = \left(\frac{20.0\text{ ly}}{0.800c}\right)\left(\frac{1\ c}{1\text{ ly/yr}}\right) = \boxed{25.0\text{ yr}}$$

(b) We see a clock on the meteoroid moving, so we do not measure proper time; that clock measures proper time.

$\Delta t = \gamma\Delta t_p$:

$$\Delta t_p = \frac{\Delta t}{\gamma} = \frac{25.0\text{ yr}}{1/\sqrt{1-v^2/c^2}} = 25.0\text{ yr}\sqrt{1-0.800^2}$$

$$= 25.0\text{ yr}(0.600) = \boxed{15.0\text{ yr}}$$

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

(c) Method one: We measure the 20.0 ly on a stick stationary in our frame, so it is proper length. The tourist measures it to be contracted to

$$L = \frac{L_p}{\gamma} = \frac{20.0 \text{ ly}}{1/\sqrt{1-0.800^2}} = \frac{20.0 \text{ ly}}{1.67} = \boxed{12.0 \text{ ly}}$$

Method two: The tourist sees the Earth approaching at $0.800c$:

$$(0.800 \text{ ly/yr})(15.0 \text{ yr}) = \boxed{12.0 \text{ ly}}$$

***P39.18** The relativistic density is

$$\frac{E_R}{c^2 V} = \frac{\gamma mc^2}{c^2 V} = \frac{\gamma m}{V} = \frac{m}{(L_p)^3\left[1-(u/c)^2\right]}$$

$$= \frac{8.00 \text{ g}}{(1.00 \text{ cm})^3\left[1-(0.900)^2\right]} = \boxed{42.1 \text{ g/cm}^3}$$

P39.19 The spaceship is measured by the Earth observer to be length-contracted to

$$L = L_p\sqrt{1-\frac{v^2}{c^2}} \qquad \text{or} \qquad L^2 = L_p^2\left(1-\frac{v^2}{c^2}\right)$$

Also, the contracted length is related to the time required to pass overhead by

$$L = v\Delta t \qquad \text{or} \qquad L^2 = v^2(\Delta t)^2 = \frac{v^2}{c^2}(c\Delta t)^2$$

Equating these two expressions gives $L_p^2 - L_p^2\frac{v^2}{c^2} = (c\Delta t)^2\frac{v^2}{c^2}$.

or $$\left[L_p^2 + (c\Delta t)^2\right]\frac{v^2}{c^2} = L_p^2$$

Using the given values $L_p = 300$ m and $\Delta t = 0.750\times10^{-6}$ s, this becomes

$$\left(1.41\times10^5 \text{ m}^2\right)\frac{v^2}{c^2} = 9.00\times10^4 \text{ m}^2$$

giving $v = \boxed{0.800c}$

P39.20 The spaceship is measured by Earth observers to be of length L, where

$$L = L_p\sqrt{1-\frac{v^2}{c^2}}$$

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

and $L = v\Delta t$

$$v\Delta t = L_p\sqrt{1-\frac{v^2}{c^2}} \qquad \text{and} \qquad v^2\Delta t^2 = L_p^2\left(1-\frac{v^2}{c^2}\right)$$

Solving for v,

$$v^2\left(\Delta t^2 + \frac{L_p^2}{c^2}\right) = L_p^2$$

giving $$\boxed{v = \frac{cL_p}{\sqrt{c^2\Delta t^2 + L_p^2}}}$$

P39.21 (a) When the source moves away from an observer, the observed frequency is

$$f' = f\left(\frac{c+v}{c-v}\right)^{1/2} = f\left(\frac{c-v_S}{c+v_S}\right)^{1/2}$$

where $v = v_{\text{source}} = -v_S$ because the source is moving away from the observer.

When $v_s \ll c$, the binomial expansion gives

$$\left(\frac{c-v_S}{c+v_S}\right)^{1/2} = \left[1-\left(\frac{v_S}{c}\right)\right]^{1/2}\left[1+\left(\frac{v_S}{c}\right)\right]^{-1/2}$$

$$\approx \left(1-\frac{v_S}{2c}\right)\left(1-\frac{v_S}{2c}\right) \approx \left(1-\frac{v_S}{c}\right)$$

So, $f' \approx f\left(1-\frac{v_S}{c}\right)$

The observed wavelength is found from $c = \lambda' f' = \lambda f$:

$$\lambda' = \frac{\lambda f}{f'} \approx \frac{\lambda f}{f\left(1-v_S/c\right)} = \frac{\lambda}{1-v_S/c}$$

$$\Delta\lambda = \lambda' - \lambda = \lambda\left(\frac{1}{1-v_S/c} - 1\right) = \lambda\left(\frac{v_S/c}{1-v_S/c}\right)$$

Since $1-\frac{v_S}{c} \approx 1$, $\boxed{\frac{\Delta\lambda}{\lambda} \approx \frac{v_S}{c}}$

(b) We use the equation from part (a) with the given values:

$$v_S = c\left(\frac{\Delta\lambda}{\lambda}\right) = c\left(\frac{20.0\text{ nm}}{397\text{ nm}}\right) = \boxed{0.050\,4c}$$

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

P39.22 We find Cooper's speed from Newton's second law:

$$\frac{GMm}{r^2} = \frac{mv^2}{r}$$

Solving,

$$v = \left[\frac{GM}{(R+h)}\right]^{1/2} = \left[\frac{\left(6.67\times10^{-11}\ \text{N}\cdot\text{m}^2/\text{kg}^2\right)\left(5.98\times10^{24}\ \text{kg}\right)}{\left(6.37\times10^{6}\ \text{m} + 0.160\times10^{6}\ \text{m}\right)}\right]^{1/2}$$

$$= 7.82\times10^{3} = 7.82\ \text{km/s}$$

Then the time period of one orbit is

$$T = \frac{2\pi(R+h)}{v} = \frac{2\pi(6.53\times10^{6}\ \text{m})}{7.82\times10^{3}\ \text{m/s}} = 5.25\times10^{3}\ \text{s}$$

(a) The time difference for 22 orbits is

$$\Delta t - \Delta t_p = (\gamma - 1)\Delta t_p = \left[\left(1 - \frac{v^2}{c^2}\right)^{-1/2} - 1\right](22T)$$

$$\Delta t - \Delta t_p \approx \left(1 + \frac{1}{2}\frac{v^2}{c^2} - 1\right)(22T) = \frac{1}{2}\left(\frac{7.82\times10^{3}\ \text{m/s}}{3.00\times10^{8}\ \text{m/s}}\right)^2$$

$$\times 22\left(5.25\times10^{3}\ \text{s}\right) = \boxed{39.2\ \mu s}$$

(b) For each one orbit Cooper aged less by

$$\Delta t - \Delta t_p = \frac{39.2\ \mu\text{s}}{22} = 1.78\ \mu\text{s}$$

The press report is $\boxed{\text{accurate to one digit}}$.

P39.23 (a) The mirror is approaching the source. Let f_m be the frequency as seen by the mirror. Thus,

$$f_m = f\sqrt{\frac{c+v}{c-v}}$$

After reflection, the mirror acts as a source, approaching the receiver. If f' is the frequency of the reflected wave,

$$f' = f_m\sqrt{\frac{c+v}{c-v}}$$

Combining gives

$$\boxed{f' = \frac{c+v}{c-v}f}$$

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

(b) Using the above result, the beat frequency is

$$f_{\text{beat}} = f' - f = f' = \frac{c+v}{c-v} f - f = f\left(\frac{c+v}{c-v} - 1\right)$$

$$f_{\text{beat}} = f\left(\frac{c+v-(c-v)}{c-v}\right) = f\left(\frac{2v}{c-v}\right) \approx f\frac{2v}{c} = \frac{2v}{c/f}$$

$$f_{\text{beat}} = \frac{2v}{\lambda}$$

(c) The wavelength is

$$\lambda = \frac{c}{f} = \frac{3.00\times10^{8}\ \text{m/s}}{10.0\times10^{9}\ \text{Hz}} = 0.030\ 0\ \text{m}$$

The beat frequency is therefore,

$$f_{\text{beat}} = \frac{2v}{\lambda} = \frac{(2)(30.0\ \text{m/s})}{(0.030\ 0\ \text{m})} = 2\ 000\ \text{Hz} = \boxed{2.00\ \text{kHz}}$$

(d) From part (b), $v = \frac{f_{\text{beat}}\lambda}{2}$, so

$$\Delta v = \frac{\Delta f_{\text{beat}}\lambda}{2} = \frac{(5.0\ \text{Hz})(0.030\ 0\ \text{m})}{2}$$

$$= \boxed{0.075\ 0\ \text{m/s} \approx 0.17\ \text{mi/h}}$$

P39.24 (a) In the Earth frame, Speedo's trip lasts for a time

$$\Delta t = \frac{\Delta x}{v} = \frac{20.0\ \text{ly}}{0.950\text{c}} = 21.05\ \text{yr}$$

Speedo's age advances only by the proper time interval

$$\Delta t_p = \frac{\Delta t}{\gamma} = 21.05\ \text{yr}\sqrt{1-(0.950)^2} = 6.574\ \text{yr}$$

during his trip. Similarly for Goslo,

$$\Delta t_p = \frac{\Delta x}{v}\sqrt{1-\frac{v^2}{c^2}} = \frac{20.0\ \text{ly}}{0.750\ \text{ly/yr}}\sqrt{1-(0.750)^2} = 17.64\ \text{yr}$$

While Speedo has landed on Planet X and is waiting for his brother, he ages by

$$\frac{20.0\ \text{ly}}{0.750\ \text{ly/yr}} - \frac{20.0\ \text{ly}}{0.950\ \text{ly/yr}} = 5.614\ \text{yr}$$

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

From their departure to when the twins meet, Speedo has aged (6.574 yr + 5.614 yr) = 12.19 yr, and Goslo has aged 17.64 years, for an age difference of

$$17.64\ \text{yr} - \left(6.574\ \text{yr} + 5.614\ \text{yr}\right) = \boxed{5.45\ \text{yr}}$$

(b) $\boxed{\text{Goslo}}$ is older.

P39.25 This problem is slightly more difficult than most, for the simple reason that your calculator probably cannot hold enough decimal places to yield an accurate answer. However, we can bypass the difficulty by noting the approximation

$$\sqrt{1 - \frac{v^2}{c^2}} \approx 1 - \frac{v^2}{2c^2}$$

Squaring both sides shows that when v/c is small, these two terms are equivalent.

We evaluate $\dfrac{v}{c} = \left(\dfrac{1\,000 \times 10^3\ \text{m/h}}{3.00 \times 10^8\ \text{m/s}}\right)\left(\dfrac{1\ \text{h}}{3\,600\ \text{s}}\right) = 9.26 \times 10^{-7}$

From Equation 39.7, the dilated time interval is

$$\Delta t = \gamma \Delta t_p = \frac{\Delta t_p}{\sqrt{1 - \dfrac{v^2}{c^2}}}$$

Rearranging, our approximation yields

$$\Delta t_p = \left(\sqrt{1 - \frac{v^2}{c^2}}\right)\Delta t \approx \left(1 - \frac{v^2}{2c^2}\right)\Delta t$$

and $$\Delta t - \Delta t_p = \frac{v^2}{2c^2}\Delta t$$

Substituting,

$$\Delta t - \Delta t_p = \frac{\left(9.26 \times 10^{-7}\right)^2}{2}(3\,600\ \text{s})$$

Thus, the time lag of the moving clock is

$$\Delta t - \Delta t_p = 1.54 \times 10^{-9}\ \text{s} = \boxed{1.54\ \text{ns}}$$

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

P39.26 The orbital speed of the Earth is as described by Newton's second law:

$$\sum F = ma: \quad \frac{Gm_S m_E}{r^2} = \frac{m_E v^2}{r}$$

Solving for the speed,

$$v = \sqrt{\frac{Gm_S}{r}} = \sqrt{\frac{\left(6.67\times10^{-11}\ \text{N}\cdot\text{m}^2/\text{kg}^2\right)\left(1.99\times10^{30}\ \text{kg}\right)}{1.496\times10^{11}\ \text{m}}}$$
$$= 2.98\times10^{4}\ \text{m/s}$$

The maximum frequency received by the extraterrestrials is

$$f'_{\max} = f\sqrt{\frac{1+v/c}{1-v/c}}$$
$$= \left(57.0\times10^{6}\ \text{Hz}\right)\sqrt{\frac{1+\left(2.98\times10^{4}\ \text{m/s}\right)/\left(3.00\times10^{8}\ \text{m/s}\right)}{1-\left(2.98\times10^{4}\ \text{m/s}\right)/\left(3.00\times10^{8}\ \text{m/s}\right)}}$$
$$= 57.005\,66\times10^{6}\ \text{Hz}$$

The minimum frequency received is

$$f'_{\min} = f\sqrt{\frac{1+v/c}{1-v/c}}$$
$$= \left(57.0\times10^{6}\ \text{Hz}\right)\sqrt{\frac{1-\left(2.98\times10^{4}\ \text{m/s}\right)/\left(3.00\times10^{8}\ \text{m/s}\right)}{1+\left(2.98\times10^{4}\ \text{m/s}\right)/\left(3.00\times10^{8}\ \text{m/s}\right)}}$$
$$= 56.994\,34\times10^{6}\ \text{Hz}$$

The difference, which allows them figure out the speed of our planet, is

$$\left(57.005\,66 - 56.994\,34\right)\times10^{6}\ \text{Hz} = \boxed{1.13\times10^{4}\ \text{Hz}}$$

Section 39.5 The Lorentz Transformation Equations

P39.27 (a) From the Lorentz transformation, the separations between the blue-light and red-light events are described by

$$\Delta x' = \gamma\left(\Delta x - v\Delta t\right):\ 0 = \gamma\left[2.00\ \text{m} - v\left(8.00\times10^{-9}\ \text{s}\right)\right]$$

$$v = \frac{2.00\ \text{m}}{8.00\times10^{-9}\ \text{s}} = \boxed{2.50\times10^{8}\ \text{m/s}}$$

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

so

$$\gamma = \frac{1}{\sqrt{1-\left(2.50\times10^{8}\ \text{m/s}\right)^{2}/\left(3.00\times10^{8}\ \text{m/s}\right)^{2}}} = 1.81$$

(b) Again from the Lorentz transformation, $x' = \gamma(x - vt)$:

$$x' = 1.81\left[3.00\ \text{m} - \left(2.50\times10^{8}\ \text{m/s}\right)\left(1.00\times10^{-9}\ \text{s}\right)\right] = \boxed{4.98\,\text{m}}$$

(c) $t' = \gamma\left(t - \frac{v}{c^2}x\right)$:

$$t' = 1.81\left[1.00\times10^{-9}\ \text{s} - \frac{\left(2.50\times10^{8}\ \text{m/s}\right)}{\left(3.00\times10^{8}\ \text{m/s}\right)^{2}}(3.00\ \text{m})\right]$$

$$t' = \boxed{-1.33\times10^{-8}\,\text{s}}$$

P39.28 Let Shannon be fixed in reference from S and see the two light-emission events with coordinates $x_1 = 0$, $t_1 = 0$, $x_2 = 0$, $t_2 = 3.00\ \mu\text{s}$. Let Kimmie be fixed in reference frame S′ and give the events coordinate $x_1' = 0$, $t_1' = 0$, $t_2' = 9.00\ \mu\text{s}$.

(a) Then we have

$$t_2' = \gamma\left(t_2 - \frac{v}{c^2}x_2\right)$$

$$9.00\ \mu\text{s} = \frac{1}{\sqrt{1 - v^2/c^2}}(3.00\ \mu\text{s} - 0)$$

$$\sqrt{1-\frac{v^2}{c^2}} = \frac{1}{3}$$

$$\boxed{v = 0.943c}$$

(b) The coordinate separation of the events is

$$\Delta x' = x_2' - x_1' = \gamma\left[(x_2 - x_1) - v(t_2 - t_1)\right]$$

$$= 3\left[0 - (0.943c)\left(3.00\times10^{-6}\ \text{s}\right)\right]\left(\frac{3.00\times10^{8}\ \text{m/s}}{c}\right)$$

$$= -2.55\times10^{3}\ \text{m}$$

$$|\Delta x'| = \boxed{2.55\times10^{3}\ \text{m}}$$

The later pulse is to the left of the origin.

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

P39.29 The rod's length perpendicular to the motion is the same in both the proper frame of the rod and in the frame in which the rod is moving–our frame:

$$\ell_y = \ell \sin\theta = \ell_{Py}$$

where ℓ_{Py} is the y component of the proper length.

We are given: $\ell = 2.00$ m , and $\theta = 30.0°,$ both measured in our reference frame. Also,

$$\gamma = \frac{1}{\sqrt{1 - v^2/c^2}} = \frac{1}{\sqrt{1 - 0.995^2}} \approx 10.0$$

As observed in our frame,

$$\ell_x = \ell \cos\theta = (2.00 \text{ m})\cos 30.0° = 1.73 \text{ m}$$

and $$\ell_y = \ell \sin\theta = (2.00 \text{ m})\sin 30.0° = 1.00 \text{ m}$$

ℓ_{Px} is a proper length, related to ℓ_x by $\ell_x = \dfrac{\ell_{Px}}{\gamma}$.

Therefore, $$\ell_{Px} = 10.0\ell_x = 17.3 \text{ m}$$

and $$\ell_{Py} = \ell_y = 1.00 \text{ m}$$

(a) $$\ell_P = \sqrt{\left(\ell_{Px}\right)^2 + \left(\ell_{Py}\right)^2} = \sqrt{\left(\frac{\ell_x}{\gamma}\right)^2 + \left(\ell_y\right)^2} = \boxed{17.4 \text{ m}}$$

(b) In the proper frame,

$$\theta_2 = \tan^{-1}\left(\frac{\ell_{Py}}{\ell_{Px}}\right) = \tan^{-1}\left(\frac{\ell_y}{\gamma \ell_x}\right) = \tan^{-1}\left(\frac{\tan 30.0°}{\gamma}\right) = \boxed{3.30°}$$

***P39.30** (a) $L_0^2 = L_{0x}^2 + L_{0y}^2$ and $L^2 = L_x^2 + L_y^2$.

Since the motion is in the x direction, the length of the rod in the y direction does not change: $L_y = L_{0y} = L_0 \sin\theta_0$ and

$$L_x = L_{0x}\sqrt{1 - \left(\frac{v}{c}\right)^2} = (L_0 \cos\theta_0)\sqrt{1 - \left(\frac{v}{c}\right)^2}$$

Thus,

$$L^2 = L_0^2 \cos^2\theta_0 \left[1 - \left(\frac{v}{c}\right)^2\right] + L_0^2 \sin^2\theta_0 = L_0^2\left[1 - \left(\frac{v}{c}\right)^2 \cos^2\theta_0\right]$$

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

or $$\boxed{L = L_0\left[1-\left(\frac{v}{c}\right)^2\cos^2\theta_0\right]^{1/2}}.$$

(b) $$\tan\theta = \frac{L_y}{L_x} = \frac{L_{0y}}{L_{0x}\sqrt{1-(v/c)^2}} = \boxed{\gamma\tan\theta_0}$$

P39.31 We use the Lorentz transformation equations 39.11. In frame S, we may take $t = 0$ for both events, so the coordinates of event A are (x = 50.0 m, $y = 0$, $z = 0$, $t = 0$), and the coordinates of event B are (x = 150 m, $y = 0$, $z = 0$, $t = 0$). The time coordinates of event A in frame S′ are

$$\begin{aligned} t'_A &= \gamma\left(t_A - \frac{v}{c^2}x_A\right) \\ &= \frac{1}{\sqrt{1-(0.800)^2}}\left(0 - \frac{0.800c}{c^2}(150\text{ m})\right) \\ &= 1.667\left(-\frac{120\text{ m}}{3.00\times10^8\text{ m/s}}\right) \\ &= -6.67\times10^{-7}\text{ s} \end{aligned}$$

The time coordinates of event B in frame S′ are

$$\begin{aligned} t'_B &= \gamma\left(t_B - \frac{v}{c^2}x_B\right) \\ &= \frac{1}{\sqrt{1-(0.800)^2}}\left(0 - \frac{0.800c}{c^2}(50.0\text{ m})\right) \\ &= 1.667\left(-\frac{40.0\text{ m}}{3.00\times10^8\text{ m/s}}\right) \\ &= -2.22\times10^{-7}\text{ s} \end{aligned}$$

We see that event B occurred earlier. The time elapsed between the events was

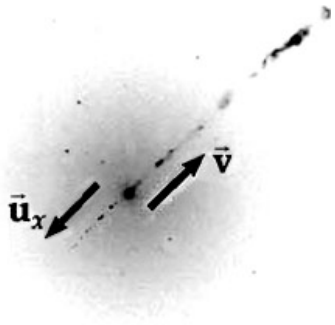
$$\Delta t' = t'_A - t'_B = \gamma\left(\Delta t - \frac{v}{c^2}\Delta x\right) = -\gamma\frac{v}{c^2}\Delta x$$

$$= -1.667\left(\frac{80.0\text{ m}}{3.00\times10^8\text{ m/s}}\right) = -4.44\times10^{-7}\text{ s} = \boxed{444\text{ ns}}$$

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

Section 39.6 The Lorentz Velocity Transformation Equations

P39.32 Take the galaxy as the unmoving frame. Arbitrarily define the jet moving upward to be the object, and the jet moving downward to be the "moving" frame:



ANS. FIG. P39.32

u_x' = velocity of other jet in frame of jet

u_x = velocity of other jet in frame of galaxy center

$= 0.750c$

v = speed of galaxy center in frame of jet $= -0.750c$

From Equation 39.16, the speed of the upward-moving jet as measured from the downward-moving jet is

$$u_x' = \frac{u_x - v}{1 - u_x v/c^2} = \frac{0.750c - (-0.750c)}{1 - (0.750c)(-0.750c)/c^2} = \frac{1.50c}{1 + 0.750^2}$$

$$= \boxed{0.960c}$$

P39.33 The question is equivalent to asking for the speed of the patrol craft in the frame of the enemy craft.

u_x' = velocity of patrol craft in frame of enemy craft

u_x = velocity of patrol craft in frame of Earth

v = speed of Earth in frame of enemy craft

From Equation 39.16,

$$u_x' = \frac{u_x - v}{1 - u_x v/c^2} = \frac{0.900c - 0.800c}{1 - (0.900)(0.800)} = \boxed{0.357c}$$

***P39.34** Let frame S be the Earth frame of reference. Then $v = -0.700c$.

The components of the velocity of the first spacecraft are

$$u_x = (0.600c)\cos 50.0° = 0.386c$$

and $$u_y = (0.600c)\sin 50.0° = 0.460c.$$

As measured from the S' frame of the second spacecraft,

$$u_x' = \frac{u_x - v}{1 - u_x v/c^2} = \frac{0.386c - (-0.700c)}{1 - \left[(0.386c)(-0.700c)/c^2\right]}$$

$$= \frac{1.086c}{1.27} = 0.855c$$

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

and

$$u_y' = \frac{u_y}{\gamma\left(1 - u_x v/c^2\right)} = \frac{0.460c\sqrt{1-(0.700)^2}}{1-(0.386)(-0.700)}$$
$$= \frac{0.460c(0.714)}{1.27} = 0.258c$$

The magnitude of $\vec{\mathbf{u}}'$ is $\sqrt{(0.855c)^2 + (0.285c)^2} = \boxed{0.893c}$

and its direction is at $\tan^{-1}\left(\frac{0.258c}{0.855c}\right) = \boxed{16.8° \text{ above the } x' \text{ axis}}$.

***P39.35** Taking to the right as positive, it is given that the velocity of the rocket relative to observer A is $v_{RA} = +0.92c$. If observer B observes the rocket to have a velocity $v_{RB} = -0.95c$, the velocity of observer B relative to the rocket is $v_{BR} = +0.95c$. The relativistic velocity addition relation then gives the velocity of B relative to the stationary observer A as

$$v_{BA} = \frac{v_{BR} + v_{RA}}{1 + \frac{v_{BR}v_{RA}}{c^2}} = \frac{+0.95c + 0.92c}{1 + \frac{(0.95c)(0.92c)}{c^2}} = +0.998c$$

or $\boxed{0.998c \text{ toward the right}}$

Section 39.7 Relativistic Linear Momentum

P39.36 (a) $p = \gamma mu$; for an electron moving at $0.010\,0c$,

$$\gamma = \frac{1}{\sqrt{1-(u/c)^2}} = \frac{1}{\sqrt{1-(0.010\,0)^2}} = 1.000\,05 \approx 1.00$$

Thus, $p = 1.00\left(9.11\times10^{-31}\text{ kg}\right)\left(0.010\,0\right)\left(3.00\times10^{8}\text{ m/s}\right)$

$$p = \boxed{2.73\times10^{-24}\text{ kg}\cdot\text{m/s}}$$

(b) Following the same steps as used in part (a), we find at $0.500c$, $\gamma = 1.15$ and

$$p = \boxed{1.58\times10^{-22}\text{ kg}\cdot\text{m/s}}$$

(c) At $0.900c$, $\gamma = 2.29$ and

$$p = \boxed{5.64\times10^{-22}\text{ kg}\cdot\text{m/s}}$$

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

P39.37 (a) The momentum condition

$$p = \gamma mu = 3mu \quad \rightarrow \quad \gamma = 3$$

From the definition of γ,

$$\gamma = \frac{1}{\sqrt{1-(u/c)^2}} \quad \rightarrow \quad u = c\sqrt{1-\frac{1}{\gamma^2}}$$

$$u = c\sqrt{1-\frac{1}{3^2}} = c\sqrt{\frac{8}{9}} = \boxed{\frac{2\sqrt{2}}{3}c = 0.943c = 2.83\times 10^8 \text{ m/s}}$$

(b) From part (a), we see the mass of the particle drops out. The result would be the same.

***P39.38** From the definition of relativistic linear momentum,

$$p = \frac{mu}{\sqrt{1-(u/c)^2}}$$

we obtain

$$1-\frac{u^2}{c^2} = \frac{m^2u^2}{p^2}$$

which gives:

$$1 = u^2\left(\frac{m^2}{p^2}+\frac{1}{c^2}\right)$$

or $\quad c^2 = u^2\left(\frac{m^2c^2}{p^2}+1\right) \quad$ and $\quad \boxed{u = \frac{c}{\sqrt{(m^2c^2/p^2)+1}}}$.

***P39.39** (a) Classically,

$$p = mv = m(0.990c) = (1.67\times 10^{-27} \text{ kg})(0.990)(3.00\times 10^8 \text{ m/s})$$

$$= \boxed{4.96\times 10^{-19} \text{ kg}\cdot\text{m/s}}$$

(b) By relativistic calculations,

$$p = \frac{mu}{\sqrt{1-(u/c)^2}} = \frac{m(0.990c)}{\sqrt{1-(0.990)^2}}$$

$$= \frac{(1.67\times 10^{-27} \text{ kg})(0.990)(3.00\times 10^8 \text{ m/s})}{\sqrt{1-(0.990)^2}}$$

$$= \boxed{3.52\times 10^{-18} \text{ kg}\cdot\text{m/s}}$$

(c) No, neglecting relativistic effects at such speeds would introduce an approximate 86% error in the result.

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

P39.40 We can express the proportion relating the speeding fine to the excess momentum as $\frac{F}{\$80.0} = \frac{\left(p_u - p_{90\ \text{km/h}}\right)}{\left(p_{190\ \text{km/h}} - p_{90\ \text{km/h}}\right)}$, where F is the fine, $p_u = \frac{mu}{\sqrt{1-(u/c)^2}}$ is the magnitude of the vehicle's momentum at speed u, and $c = 1.08 \times 10^9$ km/h. After substitution of the expression for momentum, the proportion becomes

$$\frac{F}{\$80.0} = \frac{\left[\frac{mu}{\sqrt{1-(u/c)^2}} - \frac{m(90.0\ \text{km/h})}{\sqrt{1-(90.0\ \text{km/h}/c)^2}}\right]}{\left[\frac{m(190.0\ \text{km/h})}{\sqrt{1-(190.0\ \text{km/h}/c)^2}} - \frac{m(90.0\ \text{km/h})}{\sqrt{1-(90.0\ \text{km/h}/c)^2}}\right]}$$

$$\approx \frac{\frac{u}{\sqrt{1-(u/c)^2}} - (90.0\ \text{km/h})}{100.0\ \text{km/h}}$$

(a) For $u = 1\,090$ km/h,

$$\frac{F}{\$80.0} \approx \frac{\frac{(1\,090\ \text{km/h})}{\sqrt{1-(1\,090\ \text{km/h}/1.08\times10^9\ \text{km/h})^2}} - (90.0\ \text{km/h})}{100.0\ \text{km/h}}$$

$$\approx \frac{(1\,090\ \text{km/h}) - (90.0\ \text{km/h})}{100.0\ \text{km/h}} = \frac{1\,000\ \text{km/h}}{100.0\ \text{km/h}} = 10$$

$$F = \boxed{\$800}$$

(b) For $u = 1\,000\,000\,090$ km/h,

$$\frac{F}{\$80.0} \approx \left(\frac{1}{100\ \text{km/h}}\right)$$
$$\left[\frac{(1\,000\,000\,090\ \text{km/h})}{\sqrt{1-(1\,000\,000\,090\ \text{km/h}/1.08\times10^9\ \text{km/h})^2}} - (90.0\ \text{km/h})\right]$$

$$\frac{F}{\$80.0} \approx \frac{(2.648)(1\,000\,000\,090\ \text{km/h}) - (90.0\ \text{km/h})}{100.0\ \text{km/h}}$$

$$F = \boxed{\$2.12\times10^9}$$

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

P39.41 The ratio of relativistic to classical momentum is

$$\frac{p-mu}{mu}=\frac{\gamma mu-mu}{mu}=\gamma-1$$

From the definition of γ,

$$\gamma-1=\frac{1}{\sqrt{1-(u/c)^2}}-1\approx 1+\frac{1}{2}\left(\frac{u}{c}\right)^2-1=\frac{1}{2}\left(\frac{u}{c}\right)^2$$

The ratio is then

$$\frac{p-mu}{mu}\approx\frac{1}{2}\left(\frac{90.0\text{ m/s}}{3.00\times10^8\text{ m/s}}\right)^2=\boxed{4.51\times10^{-14}}$$

P39.42 Using the relativistic form, $p=\dfrac{mu}{\sqrt{1-(u/c)^2}}=\gamma mu,$ we find the difference Δp from the classical momentum, mu:

$$\Delta p=\gamma mu-mu=(\gamma-1)mu$$

(a) The difference is 1.00% when $(\gamma-1)mu=0.010\,0\gamma mu$:

$$\gamma=\frac{1}{0.990}=\frac{1}{\sqrt{1-(u/c)^2}}$$

thus,

$$1-\left(\frac{u}{c}\right)^2=(0.990)^2,\text{ and}\qquad u=\boxed{0.141c}$$

(b) The difference is 10.0% when $(\gamma-1)mu=0.100\gamma mu$:

$$\gamma=\frac{1}{0.900}=\frac{1}{\sqrt{1-(u/c)^2}}$$

thus, $1-\left(\dfrac{u}{c}\right)^2=(0.900)^2$ and $u=\boxed{0.436c}$

P39.43 Relativistic momentum of the system of fragments must be conserved. For total momentum to be zero after as it was before, we must have, with subscript 2 referring to the heavier fragment, and subscript 1 to the lighter, $p_2=p_1$,

or $$\gamma_2m_2u_2=\gamma_1m_1u_1=\frac{2.50\times10^{-28}\text{ kg}}{\sqrt{1-(0.893)^2}}\times(0.893c)$$

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

or $$\frac{(1.67\times10^{-27}\ \text{kg})u_2}{\sqrt{1-(u_2/c)^2}} = (4.960\times10^{-28}\ \text{kg})c$$

Proceeding to solve, we find

$$\left(\frac{1.67\times10^{-27}}{4.960\times10^{-28}}\frac{u_2}{c}\right)^2 = 1-\frac{u_2^2}{c^2}$$

$$12.3\frac{u_2^2}{c^2} = 1 \quad \text{and} \quad u_2 = \boxed{0.285c}$$

Section 39.8 Relativistic Energy

***P39.44** We use the equation $\Delta E = (\gamma_1 - \gamma_2)mc^2$. For an electron, $mc^2 = 0.511$ MeV.

(a) $$\Delta E = \left(\sqrt{\frac{1}{(1-0.810)}} - \sqrt{\frac{1}{(1-0.250)}}\right)mc^2 = \boxed{0.582\ \text{MeV}}$$

(b) $$\Delta E = \left(\sqrt{\frac{1}{1-(0.990)^2}} - \sqrt{\frac{1}{1-0.810}}\right)mc^2 = \boxed{2.45\ \text{MeV}}$$

***P39.45** (a) $K = E - E_R = 5E_R$

$$E = 6E_R = 6(9.11\times10^{-31}\ \text{kg})(3.00\times10^8\ \text{m/s})^2 = 4.92\times10^{-13}\ \text{J}$$
$$= \boxed{3.07\ \text{MeV}}$$

(b) $E = \gamma mc^2 = \gamma E_R$

Thus, $\gamma = \dfrac{E}{E_R} = 6 = \dfrac{1}{\sqrt{1-u^2/c^2}}$ which yields $\boxed{u = 0.986c}$

P39.46 (a) To find the speed of the protons with $E = \gamma mc^2 = 400mc^2$, we write

$$\gamma = \frac{1}{\sqrt{1-(u/c)^2}} \quad \rightarrow \quad u = c\sqrt{1-\frac{1}{\gamma^2}}$$

So, $$u = c\sqrt{1-\frac{1}{(400)^2}} = \boxed{0.999\,997c}$$

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

(b) From Example 39.9, for a proton, $mc^2 = 938$ MeV. Then

$$K = (\gamma - 1)mc^2 = 399(938 \text{ MeV}) = \boxed{3.74 \times 10^5 \text{ MeV}}$$

P39.47 At $u = 0.950c$, it will be useful to know the gamma factor:

$$\gamma = \frac{1}{\sqrt{1 - u^2/c^2}} = \frac{1}{\sqrt{1 - 0.950^2}} = 3.20$$

(a) The rest energy is

$$E_R = mc^2 = (1.67 \times 10^{-27} \text{ kg})(2.998 \times 10^8 \text{ m/s})^2$$
$$= 1.50 \times 10^{-10} \text{ J}$$
$$= 1.50 \times 10^{-10} \text{ J}\left(\frac{1 \text{ eV}}{1.60 \times 10^{-19} \text{ J}}\right) = \boxed{938 \text{ MeV}}$$

(We use a value for c accurate to four digits so that we can be sure to get an answer accurate to three digits. Through the rest of the book we will use values for physical constants accurate to four digits or to three, whichever we like. We will still quote answers to three digits, and you can still think of the last digit as uncertain.)

(b) The total energy is

$$E = \gamma mc^2 = \gamma E_R = (3.20)(938 \text{ MeV}) = \boxed{3.00 \text{ GeV}}$$

(c) The kinetic energy is

$$K = E - E_R = 3.00 \text{ GeV} - 938 \text{ MeV} = \boxed{2.07 \text{ GeV}}$$

P39.48 (a) Using the classical equation,

$$K = \frac{1}{2}mu^2 = \frac{1}{2}(78.0 \text{ kg})(1.06 \times 10^5 \text{ m/s})^2 = \boxed{4.38 \times 10^{11} \text{ J}}$$

(b) Using the relativistic equation, $K = \left(\frac{1}{\sqrt{1-(u/c)^2}} - 1\right)mc^2$:

$$K = \left[\frac{1}{\sqrt{1-\left(\frac{1.06 \times 10^5}{2.998 \times 10^8}\right)^2}} - 1\right](78.0 \text{ kg})(3.00 \times 10^8 \text{ m/s})^2$$
$$= \boxed{4.38 \times 10^{11} \text{J}}$$

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

(c) When $\frac{u}{c} \ll 1$, the binomial series expansion gives

$$\left[1-\left(\frac{u}{c}\right)^2\right]^{-1/2} \approx 1+\frac{1}{2}\left(\frac{u}{c}\right)^2$$

Thus, $\left[1-\left(\frac{u}{c}\right)^2\right]^{-1/2} - 1 \approx \frac{1}{2}\left(\frac{u}{c}\right)^2$ and the relativistic expression for kinetic energy becomes $K \approx \frac{1}{2}\left(\frac{u}{c}\right)^2 mc^2 = \frac{1}{2}mu^2$. That is, in the limit of speeds much smaller than the speed of light, the relativistic and classical expressions yield the same results. In this situation the two kinetic energy values are experimentally indistinguishable. The fastest-moving macroscopic objects launched by human beings move sufficiently slowly compared to light that relativistic corrections to their energy are negligible.

P39.49 The work–kinetic energy theorem is $W = \Delta K = K_f - K_i$, which for relativistic speeds (u comparable to c) is:

$$W = \left(\frac{1}{\sqrt{1-u_f^2/c^2}} - 1\right)mc^2 - \left(\frac{1}{\sqrt{1-u_i^2/c^2}} - 1\right)mc^2$$

or, simplified,

$$W = \left(1/\sqrt{1-u_f^2/c^2} - 1/\sqrt{1-u_i^2/c^2}\right)mc^2$$

From our specialized equation,

(a) $$W = \left(\frac{1}{\sqrt{1-0.750^2}} - \frac{1}{\sqrt{1-0.500^2}}\right) \times \left(1.67\times10^{-27}\text{ kg}\right)\left(3.00\times10^{8}\text{ m/s}\right)^2$$

$$W = (1.512 - 1.155)\left(1.50\times10^{-10}\text{ J}\right) = \boxed{5.37\times10^{-11}\text{ J}} = 336\text{ MeV}$$

(b) $$W = \left(\frac{1}{\sqrt{1-0.995^2}} - \frac{1}{\sqrt{1-0.500^2}}\right) \left(1.67\times10^{-27}\text{ kg}\right)\left(3.00\times10^{8}\text{ m/s}\right)^2$$

$$W = \left(10.01 - 1.155\right)\left(1.50\times10^{-10}\text{ J}\right) = \boxed{1.33\times10^{-9}\text{ J}} = 8.32\text{ GeV}$$

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

P39.50 The relativistic kinetic energy of an object of mass m and speed u is $K_r = \left(\frac{1}{\sqrt{1-u^2/c^2}} - 1 \right) mc^2$. The classical equation is $K_c = \frac{1}{2}mu^2$. Their ratio is

$$\frac{K_r}{K_c} = \frac{\left(\frac{1}{\sqrt{1-u^2/c^2}} - 1 \right) mc^2}{\frac{1}{2}mu^2} = \frac{2\left(\frac{1}{\sqrt{1-u^2/c^2}} - 1 \right)}{u^2/c^2}$$

$$= 2\left(\frac{1}{\sqrt{1-u^2/c^2}} - 1 \right) \frac{1}{u^2/c^2}$$

$$\frac{K_r}{K_c} = 2\left(\frac{1}{\sqrt{1-(0.100)^2}} - 1 \right) \frac{1}{(0.100)^2} = 1.007\ 56$$

For still smaller speeds the agreement will be still better.

P39.51 Given $E = 2mc^2$, where $mc^2 = 938$ MeV from Example 39.9. We use Equation 39.27:

$$E^2 = p^2c^2 + \left(mc^2\right)^2$$

$$\left(2mc^2\right)^2 = p^2c^2 + \left(mc^2\right)^2$$

$$4\left(mc^2\right)^2 = p^2c^2 + \left(mc^2\right)^2 \quad \rightarrow \quad p^2c^2 = 3\left(mc^2\right)^2$$

Solving for the momentum then gives

$$p = \sqrt{3}\frac{\left(mc^2\right)}{c} = \sqrt{3}\frac{(938\ \text{MeV})}{c} = \boxed{1.62 \times 10^3\ \text{MeV}/c}$$

P39.52 (a) $E = \gamma mc^2 = 20.0$ GeV with $mc^2 = 0.511$ MeV for electrons.

Thus, $\gamma = \frac{20.0 \times 10^9\ \text{eV}}{0.511 \times 10^6\ \text{eV}} = \boxed{3.91 \times 10^4}$.

(b) $\gamma = \frac{1}{\sqrt{1-\left(u/c\right)^2}} \quad \rightarrow \quad u = c\sqrt{1 - \frac{1}{\gamma^2}} = \boxed{0.999\ 999\ 999\ 7c}$

(c) $L = L_p\sqrt{1-\left(\frac{u}{c}\right)^2} = \frac{L_p}{\gamma} = \frac{3.00 \times 10^3\ \text{m}}{3.91 \times 10^4} = 7.67 \times 10^{-2}\ \text{m} = \boxed{7.67\ \text{cm}}$

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

P39.53 (a) $E = 2.86 \times 10^5 \text{ J}$ leaves the system, so the final mass is $\boxed{\text{smaller}}$.

(b) The mass-energy relation says that $E = mc^2$. Therefore,

$$m = \frac{E}{c^2} = \frac{2.86 \times 10^5 \text{ J}}{\left(3.00 \times 10^8 \text{ m/s}\right)^2} = \boxed{3.18 \times 10^{-12} \text{ kg}}$$

(c) $\boxed{\text{It is too small a fraction of 9.00 g to be measured}}$.

P39.54 The loss of mass in the nuclear reactor is

$$\Delta m = \frac{E}{c^2} = \frac{P\,\Delta t}{c^2}$$

$$= \frac{0.800\left(1.00 \times 10^9 \text{ J/s}\right)\left(3.00 \text{ yr}\right)\left(3.16 \times 10^7 \text{ s/yr}\right)}{\left(3.00 \times 10^8 \text{ m/s}\right)^2}$$

$$= \boxed{0.842 \text{ kg}}$$

P39.55 The power output of the Sun is

$$P = \frac{dE}{dt} = \frac{d\left(mc^2\right)}{dt} = c^2 \frac{dm}{dt} = 3.85 \times 10^{26} \text{ W}$$

Thus, $$\frac{dm}{dt} = \frac{3.85 \times 10^{26} \text{ J/s}}{\left(3.00 \times 10^8 \text{ m/s}\right)^2} = \boxed{4.28 \times 10^9 \text{ kg/s}}$$

P39.56 Total energy is conserved. The photon must have enough energy to be able to create an electron and a positron, both having the same rest mass:

$$E_\gamma \geq 2m_e c^2 = 1.02 \text{ MeV} \quad \rightarrow \quad E_\gamma \geq \boxed{1.02 \text{ MeV}}$$

P39.57 We use Equation 39.23 for relativistic kinetic energy.

(a) The change in kinetic energy of the spaceship is the minimum energy required to accelerate the spaceship. From Equation 39.23, relativistic kinetic energy is given by

$$K = \left(\gamma - 1\right)mc^2 = \left(\frac{1}{\sqrt{1 - u^2/c^2}} - 1\right)mc^2$$

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

The change in kinetic energy is then

$$\Delta K = \left(\gamma_f - 1\right)mc^2 - \left(\gamma_i - 1\right)mc^2 = \left(\gamma_f - \gamma_i\right)mc^2$$

$$= \left(\frac{1}{\sqrt{1 - u_f^2/c^2}} - \frac{1}{\sqrt{1 - u_i^2/c^2}} \right) mc^2$$

$$= \left(\frac{1}{\sqrt{1 - (0.700)^2}} - \frac{1}{\sqrt{1 - 0}} \right) mc^2$$

$$= (1.40 - 1)\left(2.40 \times 10^6 \text{ kg}\right)\left(2.998 \times 10^8 \text{ m/s}\right)^2$$

$$= \boxed{8.63 \times 10^{22} \text{ J}}$$

(b) We use Einstein's famous mass-energy relation, and equate the rest energy of the fuel to the change in kinetic energy of the spacecraft:

$$E = mc^2 = \Delta K$$

The required mass of fuel is then

$$m = \frac{\Delta K}{c^2} = \left(\frac{1}{\sqrt{1 - (0.700)^2}} - 1 \right)\left(2.40 \times 10^6 \text{ kg}\right) = \boxed{9.61 \times 10^5 \text{ kg}}$$

P39.58 We are told to start from $E = \gamma mc^2$ and $p = \gamma mu$. Squaring both equations gives

$$E^2 = \left(\gamma mc^2\right)^2 \qquad \text{and} \qquad p^2 = \left(\gamma mu\right)^2$$

We choose to multiply the second equation by c^2 and subtract it from the first:

$$E^2 - p^2c^2 = \left(\gamma mc^2\right)^2 - (\gamma mu)^2 c^2$$

We factor to obtain

$$E^2 - p^2c^2 = \gamma^2\left[\left(mc^2\right)\left(mc^2\right) - \left(mc^2\right)\left(mu^2\right)\right]$$

Extracting the (mc^2) factors gives

$$E^2 - p^2c^2 = \gamma^2\left(mc^2\right)^2\left(1 - \frac{u^2}{c^2}\right)$$

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

We substitute the definition of γ:

$$E^2 - p^2c^2 = \left(1 - \frac{u^2}{c^2}\right)^{-1}\left(mc^2\right)^2\left(1 - \frac{u^2}{c^2}\right)$$

The γ^2 factors divide out, leaving

$$E^2 - p^2c^2 = \left(mc^2\right)^2$$

P39.59 From $K = (\gamma - 1)mc^2 = \left(\frac{1}{\sqrt{1 - u^2/c^2}} - 1\right)mc^2$, we have

$$\frac{K}{mc^2} + 1 = \frac{1}{\sqrt{1 - u^2/c^2}} = \frac{K + mc^2}{mc^2}$$

$$1 - \frac{u^2}{c^2} = \frac{m^2c^4}{\left(K + mc^2\right)^2}$$

$$\frac{u^2}{c^2} = 1 - \frac{\left(mc^2\right)^2}{\left(K + mc^2\right)^2}$$

$$u = c\left[1 - \left(\frac{mc^2}{K + mc^2}\right)^2\right]^{1/2}$$

(a) Electron: $u = c\left[1 - \left(\frac{0.511}{2.511}\right)^2\right]^{1/2} = \boxed{0.979c}$

(b) Proton: $u = c\left[1 - \left(\frac{938}{940}\right)^2\right]^{1/2} = \boxed{0.065\,2c}$

(c) $\frac{u_{\text{electron}}}{u_{\text{proton}}} = \frac{0.979c}{0.065\,2c} = \boxed{15.0}$

In this case the electron is moving relativistically, but the classical expression $\frac{1}{2}mv^2$ is accurate to two digits for the proton.

(d) Electron: $u = c\left[1 - \left(\frac{0.511}{2\,000.511}\right)^2\right]^{1/2} = \boxed{0.999\,999\,97c}$

Proton: $u = c\left[1 - \left(\frac{938}{2\,938}\right)^2\right]^{1/2} = \boxed{0.948c}$

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

Then,

$$\frac{u_{\text{electron}}}{u_{\text{proton}}} = \frac{c\left[1-\left(\dfrac{0.511}{2\,000.511}\right)^2\right]^{1/2}}{c\left[1-\left(\dfrac{938}{2\,938}\right)^2\right]^{1/2}} = \boxed{1.06}$$

As the kinetic energies of both particles become large, their speeds approach c. By contrast, classically the speed would become large without any finite limit.

P39.60 The kinetic energy of the car is given by

$$K = (\gamma - 1)mc^2 = \left(\left(1 - u^2/c^2\right)^{-1/2} - 1\right)mc^2$$

We use the series expansion from Appendix B.5:

$$K = mc^2\left[1 + \left(-\frac{1}{2}\right)(-u^2/c^2) + \left(-\frac{1}{2}\right)\left(-\frac{3}{2}\right)\frac{1}{2}(-u^2/c^2)^2 + \cdots - 1\right]$$

$$K = \frac{1}{2}mu^2 + \frac{3}{8}m\frac{u^4}{c^2} + \cdots$$

The actual kinetic energy, given by this relativistic equation, is $\boxed{\text{larger}}$ than the classical $\frac{1}{2}mu^2$.

The difference, for $m = 1\,000$ kg and $u = 25$ m/s, is

$$\frac{3}{8}m\frac{u^4}{c^2} = \frac{3}{8}(1\,000\text{ kg})\frac{(25\text{ m/s})^4}{(3.00\times10^8\text{ m/s})^2} = 1.6\times10^{-9}\text{ J}\ \boxed{\sim 10^{-9}\text{ J}}$$

P39.61 We use, together, both the energy version and the momentum version of the isolated system model. By conservation of system energy,

$$m_\pi c^2 = \gamma m_\mu c^2 + |p_{\bar{\nu}}|c$$

By conservation of system momentum:

$$p_{\bar{\nu}} = -p_\mu = -\gamma m_\mu u$$

Substituting the second equation into the first,

$$m_\pi c^2 = \gamma m_\mu c^2 + \gamma m_\mu uc$$

Simplified, this equation then reads

$$m_\pi = m_\mu(\gamma + \gamma u/c)$$

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

Substituting the masses,

$$273m_e = (207m_e)(\gamma + \gamma u/c)$$

where the rest energy of an electron is

$$m_e c^2 = 0.511 \text{ MeV}$$

Numerically,

$$\frac{273m_e}{207m_e} = \frac{1 + u/c}{\sqrt{1 - (u/c)^2}} = \sqrt{\frac{1 + u/c}{1 - u/c}}$$

Solving for the muon speed,

$$\frac{u}{c} = \frac{273^2 - 207^2}{273^2 + 207^2} = 0.270$$

Therefore,

$$\gamma = \frac{1}{\sqrt{1 - u^2/c^2}} = 1.038\ 5$$

(a) and the muon's kinetic energy is

$$K_\mu = (0.038\ 5)(207 \times 0.511 \text{ MeV}) = \boxed{4.08 \text{ MeV}}$$

(b) The energy of the antineutrino is

$$K_{\bar{\nu}} = (273 \times 0.511 \text{ MeV}) - (207 \times 0.511 \text{ MeV} + 4.08 \text{ MeV})$$
$$= \boxed{29.6 \text{ MeV}}$$

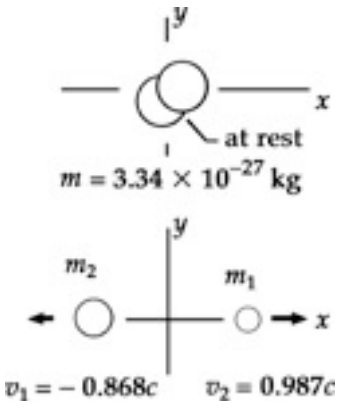
P39.62 (a) The initial system is $\boxed{\text{isolated}}$.

(b) $\boxed{\text{Isolated system: conservation of energy, and isolated system: conservation of momentum.}}$

(c) We must conserve both energy and relativistic momentum of the system of fragments. With subscript 1 referring to the 0.987c particle and subscript 2 to the 0.868c particle,

$$\gamma_1 = \frac{1}{\sqrt{1 - (0.987)^2}} = \boxed{6.22} \text{ and}$$

$$\gamma_2 = \frac{1}{\sqrt{1 - (0.868)^2}} = \boxed{2.01}$$



ANS. FIG. P39.62

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

(d) Conservation of energy gives $E_1 + E_2 = E_{\text{total}}$

which is

$$\gamma_1 m_1 c^2 + \gamma_2 m_2 c^2 = m_{\text{total}} c^2$$

or $6.22m_1 + 2.01m_2 = 3.34 \times 10^{-27}$ kg

This reduces to: $\boxed{3.09m_1 + m_2 = 1.66 \times 10^{-27} \text{ kg}}$ **[1]**

(e) Since the final momentum of the system must equal zero, $p_1 = p_2$ gives

$$\gamma_1 m_1 u_1 = \gamma_2 m_2 u_2$$

or $(6.22)(0.987c)m_1 = (2.01)(0.868c)m_2$

which becomes $\boxed{m_2 = 3.52m_1}$ **[2]**

(f) Substituting [2] into [1] gives

$$3.09m_1 + 3.52m_1 = 1.66 \times 10^{-27} \text{ kg}$$

thus, $m_1 = \boxed{2.51 \times 10^{-28} \text{ kg}}$ and $m_2 = \boxed{8.84 \times 10^{-28} \text{ kg}}$

P39.63 Let $m = 1.99 \times 10^{-26}$ kg, and $\vec{\mathbf{u}} = u\hat{\mathbf{i}} = 0.500c\hat{\mathbf{i}}$. An isolated system of two particles of mass m and $m' = m/3$ collide with the respective velocities $\vec{\mathbf{u}}$ and $-\vec{\mathbf{u}}$, resulting in a particle with mass M and velocity $\vec{\mathbf{v}}_f = v_f\hat{\mathbf{i}}$.
By conservation of the x component of momentum (γmu):

$$\frac{mu}{\sqrt{1-u^2/c^2}} + \frac{m(-u)}{3\sqrt{1-u^2/c^2}} = \frac{Mv_f}{\sqrt{1-v_f^2/c^2}}$$

$$\frac{2mu}{3\sqrt{1-u^2/c^2}} = \frac{Mv_f}{\sqrt{1-v_f^2/c^2}} \quad \textbf{[1]}$$

By conservation of total energy (γmc^2):

$$\frac{mc^2}{\sqrt{1-u^2/c^2}} + \frac{mc^2}{3\sqrt{1-u^2/c^2}} = \frac{Mc^2}{\sqrt{1-v_f^2/c^2}}$$

$$\frac{4mc^2}{3\sqrt{1-u^2/c^2}} = \frac{Mc^2}{\sqrt{1-v_f^2/c^2}} \quad \textbf{[2]}$$

To start solving, we divide the momentum equation [1] by the energy equation [2], giving $v_f = \frac{2u}{4} = \frac{u}{2}$.

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

Then, substituting the value of the final speed back into the energy equation [2], we get

$$\frac{Mc^2}{\sqrt{1-u^2/4c^2}} = \frac{4mc^2}{3\sqrt{1-u^2/c^2}}$$

$$\frac{2Mc^2}{\sqrt{4-u^2/c^2}} = \frac{4mc^2}{3\sqrt{1-u^2/c^2}}$$

$$M = \frac{2m\sqrt{4-u^2/c^2}}{3\sqrt{1-u^2/c^2}} = \frac{2\left(1.99\times10^{-26}\text{ kg}\right)\sqrt{4-(0.500)^2}}{3\sqrt{1-(0.500)^2}}$$

$$M = \boxed{2.97\times10^{-26}\text{ kg}}$$

P39.64 (a) By conservation of the x component of momentum (γmu):

$$\frac{mu}{\sqrt{1-u^2/c^2}} + \frac{m(-u)}{3\sqrt{1-u^2/c^2}} = \frac{Mv_f}{\sqrt{1-v_f^2/c^2}}$$

$$\frac{2mu}{3\sqrt{1-u^2/c^2}} = \frac{Mv_f}{\sqrt{1-v_f^2/c^2}} \qquad \textbf{[1]}$$

By conservation of total energy (γmc^2):

$$\frac{mc^2}{\sqrt{1-u^2/c^2}} + \frac{mc^2}{3\sqrt{1-u^2/c^2}} = \frac{Mc^2}{\sqrt{1-v_f^2/c^2}}$$

$$\frac{4mc^2}{3\sqrt{1-u^2/c^2}} = \frac{Mc^2}{\sqrt{1-v_f^2/c^2}} \qquad \textbf{[2]}$$

To start solving, we divide the momentum equation [1] by the energy equation [2], giving $v_f = \frac{2u}{4} = \frac{u}{2}$. Then, substituting the value of the final speed back into the energy equation [2], we get

$$\frac{Mc^2}{\sqrt{1-u^2/4c^2}} = \frac{4mc^2}{3\sqrt{1-u^2/c^2}}$$

$$\frac{2Mc^2}{\sqrt{4-u^2/c^2}} = \frac{4mc^2}{3\sqrt{1-u^2/c^2}}$$

$$\boxed{M = \frac{2m\sqrt{4-u^2/c^2}}{3\sqrt{1-u^2/c^2}}}$$

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

(b) As $u \to 0$, $M = \dfrac{2m\sqrt{4 - u^2/c^2}}{3\sqrt{1 - u^2/c^2}} \to \dfrac{2m\sqrt{4}}{3\sqrt{1}} = \boxed{\dfrac{4m}{3}}$

(c) The answer to part (b) is in agreement with the classical result, which is the arithmetic sum of the masses of the two colliding particles.

Section 39.9 The General Theory of Relativity

P39.65 (a) For the satellite, Newton's second law gives

$$\sum F = ma: \quad \frac{GM_E m}{r^2} = \frac{mv^2}{r} = \frac{m}{r}\left(\frac{2\pi r}{T}\right)^2$$

which gives

$$GM_E T^2 = 4\pi^2 r^3$$

Solving for the orbital radius,

$$r = \left(\frac{GM_E T^2}{4\pi^2}\right)^{1/3}$$

$$r = \left[\frac{\left(6.67 \times 10^{-11}\ \text{N} \cdot \text{m}^2/\text{kg}^2\right)\left(5.98 \times 10^{24}\ \text{kg}\right)\left(43\,080\ \text{s}\right)^2}{4\pi^2}\right]^{1/3}$$

$$= \boxed{2.66 \times 10^7\ \text{m}}$$

(b) $$v = \frac{2\pi r}{T} = \frac{2\pi\left(2.66 \times 10^7\ \text{m}\right)}{43\,080\ \text{s}} = \boxed{3.87 \times 10^3\ \text{m/s}}$$

(c) From the relationship of frequency and period:

$$f = \frac{1}{T} \quad \to \quad df = -\frac{dT}{T^2} = -f\left(\frac{dT}{T}\right) \quad \to \quad \frac{df}{f} = -\frac{dT}{T}$$

We see the fractional decrease in frequency is equal in magnitude to the fractional change in period.

The small fractional decrease in frequency received is equal in magnitude to the fractional increase in period of the moving oscillator due to time dilation:

$$\frac{df}{f} = -\frac{dT}{T} = -\frac{\gamma\Delta t_P - \Delta t_P}{\Delta t_P} = -(\gamma - 1)$$

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

$$\frac{df}{f} = -\left(\frac{1}{\sqrt{1-(v/c)^2}} - 1\right) = 1 - \frac{1}{\sqrt{1-(v/c)^2}}$$

$$\approx 1 - \left[1 + \frac{1}{2}\left(\frac{v}{c}\right)^2\right] = -\frac{1}{2}\left(\frac{v}{c}\right)^2$$

$$= -\frac{1}{2}\left[\left(\frac{3.87\times10^3 \text{ m/s}}{3.00\times10^8 \text{ m/s}}\right)^2\right] = \boxed{-8.34\times10^{-11}}$$

(d) The orbit altitude is large compared to the radius of the Earth, so we must use

$$U_g = -\frac{GM_E m}{r}$$

The change in gravitational potential energy is

$$\Delta U_g = -\left(6.67\times10^{-11}\ \frac{\text{N}\cdot\text{m}^2}{\text{kg}^2}\right) \times (5.98\times10^{24}\ \text{kg})m\left[\frac{1}{2.66\times10^7\ \text{m}} - \frac{1}{6.37\times10^6\ \text{m}}\right]$$

$$= (4.76\times10^7\ \text{J/kg})m$$

Then

$$\frac{\Delta f}{f} = \frac{\Delta U_g}{mc^2} = \frac{(4.76\times10^7\ \text{J/kg})\cancel{m}}{\cancel{m}(3.00\times10^8\ \text{m/s})^2} = \boxed{+5.29\times10^{-10}}$$

(e) $-8.34\times10^{-11} + 5.29\times10^{-10} = \boxed{+4.46\times10^{-10}}$

Additional Problems

P39.66 (a) When $K_e = K_p$, $m_e c^2(\gamma_e - 1) = m_p c^2(\gamma_p - 1)$

In this case, $m_e c^2 = 0.511$ MeV, $m_p c^2 = 938$ MeV

and $\gamma_e = \left[1 - (0.750)^2\right]^{-1/2} = 1.511\,9$

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

Substituting, $\gamma_p = 1 + \dfrac{m_e c^2 (\gamma_e - 1)}{m_p c^2} = 1 + \dfrac{(0.511 \text{ MeV})(1.511\,9 - 1)}{938 \text{ MeV}}$

$$= 1.000\,279$$

But $$\gamma_p = \frac{1}{\left[1 - \left(u_p / c\right)^2\right]^{1/2}}$$

Therefore, $u_p = c\sqrt{1 - \gamma_p^{-2}} = \boxed{0.023\ 6c}$

(b) When $p_e = p_p$, $\gamma_p m_p u_p = \gamma_e m_e u_e$ or $\gamma_p u_p = \dfrac{\gamma_e m_e u_e}{m_p}$

Thus, $\gamma_p u_p = \dfrac{(1.511\,9)(0.511 \text{ MeV}/c^2)(0.750c)}{938 \text{ MeV}/c^2} = 6.177\,2 \times 10^{-4} c$

and $\dfrac{u_p}{c} = 6.177\,2 \times 10^{-4} \sqrt{1 - \left(\dfrac{u_p}{c}\right)^2}$

which yields $u_p = \boxed{6.18 \times 10^{-4} c} = 185 \text{ km/s}$

P39.67 The original rest energy of four protons is

$$E_R = 4(938.78 \text{ MeV}) = 3\,755.12 \text{ MeV}$$

The energy given off is

$$|\Delta E| = (3\,755.12 - 3\,728.4) \text{ MeV} = 26.7 \text{ MeV}$$

The fractional energy released is

$$\frac{|\Delta E|}{E_R} = \frac{26.7 \text{ MeV}}{3\,755 \text{ MeV}} \times 100\% = \boxed{0.712\%}$$

P39.68 From the particle under constant speed model, find the travel time for Speedo from Goslo's reference frame:

$$\Delta t = \frac{d}{u} = \frac{2(50 \text{ ly})}{0.85c}\left(\frac{c \cdot \text{yr}}{\text{ly}}\right) = 118 \text{ yr}$$

Therefore, when Speedo arrives back on Earth, 118 years have passed and Goslo would have to be 158 years old. Furthermore, Speedo will be 102 years old. Perhaps future medical breakthroughs may extend the life expectancy to 158 years and beyond, but that is impossible at present.

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

***P39.69** (a) Consider the raindrops moving toward the station, at speed v. They receive radio waves with the Doppler-enhanced frequency $f' = f\sqrt{\frac{c+v}{c-v}}$, where $f = 2.85$ GHz. These raindrops reflect the waves at frequency f'. The waves are received by the station with another upward Doppler shift in frequency to

$$f'' = f'\sqrt{\frac{c+v}{c-v}} = f\sqrt{\frac{c+v}{c-v}}\sqrt{\frac{c+v}{c-v}}$$

$$2.85\times10^{9}\ \text{Hz} + 254\ \text{Hz} = 2.85\times10^{9}\ \text{Hz}\left(\frac{c+v}{c-v}\right)$$

$$1 + 8.91\times10^{-8} = \frac{c+v}{c-v}$$

$$c + 8.91\times10^{-8}c - v - 8.91\times10^{-8}v = c + v$$

$$8.91\times10^{-8}c = 2.000\,000\,089v$$

$$v = \left(4.46\times10^{-8}\right)\left(3.00\times10^{8}\ \text{m/s}\right) = \boxed{13.4\ \text{m/s}}$$

The same calculation with 254 Hz replaced by –254 Hz applies to the receding raindrops and given the same velocity magnitude. Thus the velocities are 13.4 m/s toward the station and 13.4 m/s away from the station.

(b) Radio waves travel to the rain and back again in $180\ \mu\text{s}$, so the one-way distance is $\frac{1}{2}\left(3.00\times10^{8}\ \text{m/s}\right)\left(180\times10^{-6}\ \text{s}\right) = 27\,000$ m. The frequency shifts indicate the batch of raindrops are whirling around a common center separated by 1° of arc. The diameter of the vortex is

$$s = r\theta = (27\,000\ \text{m})(1°)\left(\frac{\pi\ \text{rad}}{180°}\right) = 471\ \text{m}$$

Its radius is therefore $\frac{1}{2}(471\ \text{m}) = 236\ \text{m}$ and the angular speed of the rain is $\omega = \frac{v}{r} = \frac{13.4\ \text{m/s}}{236\ \text{m}} = \boxed{0.056\,7\ \text{rad/s}}$. A Doppler weather radar computer performs a calculation like this to detect a "tornado vortex signature."

***P39.70** From energy conservation, we have

$$\frac{(1\,400\ \text{kg})c^2}{\sqrt{1-0^2}} + \frac{(900\ \text{kg})c^2}{\sqrt{1-0.850^2}} = \frac{Mc^2}{\sqrt{1-v^2/c^2}}$$

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

$$(3\,108\text{ kg})\sqrt{1-\frac{v^2}{c^2}} = M$$

From momentum conservation, we have

$$0+\frac{(900\text{ kg})(0.850c)}{\sqrt{1-0.850^2}} = \frac{Mv}{\sqrt{1-v^2/c^2}}$$

$$(1\,452\text{ kg})\sqrt{1-\frac{v^2}{c^2}} = \frac{Mv}{c}$$

(a) Dividing the momentum equation by the energy equation gives

$$\frac{v}{c} = \frac{1\,452}{3\,108} = 0.467\,, \quad \text{or} \qquad \boxed{v = 0.467c}$$

(b) Now by substitution, $(3\,108\text{ kg})\sqrt{1-0.467^2} = \boxed{M = 2.75\times10^3\text{ kg}}$.

***P39.71** (a) Observers on Earth measure the distance to Andromeda to be

$$d = 2.00\times10^6\text{ ly} = \left(2.00\times10^6\text{ ly}\right)c$$

The time for the trip, in Earth's frame of reference, is

$$\Delta t = \gamma\Delta t_p = \frac{30.0\text{ yr}}{\sqrt{1-(v/c)^2}}$$

The required speed is then

$$v = \frac{d}{\Delta t} = \frac{\left(2.00\times10^6\text{ ly}\right)c}{(30.0\text{ yr})/\sqrt{1-(v/c)^2}}$$

which gives, suppressing units,

$$\left(1.50\times10^{-5}\right)(v/c) = \sqrt{1-(v/c)^2}$$

Squaring both sides of this equation and solving for v/c yields

$$\frac{v}{c} = \frac{1}{\sqrt{1+2.25\times10^{-10}}}$$

Then, the approximation $\frac{1}{\sqrt{1+x}} = 1-\frac{x}{2}$ gives

$$\frac{v}{c} = 1-\frac{2.25\times10^{-10}}{2} = \boxed{1-1.12\times10^{-10}}$$

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

(b) Let $\frac{v}{c} = \frac{1}{\sqrt{1+x}}$, where $x = 2.25 \times 10^{-10}$.. Then,

$$1 - \left(\frac{v}{c}\right)^2 = 1 - \frac{1}{1+x} = \frac{x}{1+x}$$

and

$$\gamma = \frac{1}{\sqrt{1-\left(\frac{v}{c}\right)^2}} = \sqrt{\frac{1+x}{x}} = \sqrt{1+\frac{1}{x}}$$

The kinetic energy of the spacecraft is given by

$$KE = (\gamma - 1)mc^2 = \left(\sqrt{1+\frac{1}{x}} - 1\right)mc^2$$

Thus,

$$KE = \left(\sqrt{1+\frac{1}{2.25\times10^{-10}}} - 1\right)\left(1.00\times10^6 \text{ kg}\right)\left(3.00\times10^8 \text{ m/s}\right)^2$$

$$= 5.99991\times10^{27} \text{ J} = \boxed{6.00\times10^{27} \text{ J}}$$

(c) The cost of this energy is

$$\text{cost} = KE \times \text{rate} = \left[\left(6.00\times10^{27} \text{ J}\right)\left(\frac{1 \text{ kWh}}{3.60\times10^6 \text{ J}}\right)\right](\$0.13/\text{kWh})$$

$$= \boxed{\$2.17\times10^{20}}$$

***P39.72** In this case, the proper time is T_0 (the time measured by the students on a clock at rest relative to them). The dilated time measured by the professor is:

$$\Delta t = \gamma T_0$$

where $\Delta t = T + t$. Here T is the time she waits before sending a signal and t is the time required for the signal to reach the students. Thus, we have:

$$T + t = \gamma T_0 \qquad \textbf{[1]}$$

To determine the travel time t, realize that the distance the students will have moved beyond the professor before the signal reaches them is:

$$d = v(T + t)$$

The time required for the signal to travel this distance is:

$$t = \frac{d}{c} = \left(\frac{v}{c}\right)(T + t)$$

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

Solving for t gives:

$$t = \frac{(v/c)T}{1-(v/c)}$$

Substituting this into equation [1] yields:

$$T + \frac{(v/c)T}{1-(v/c)} = \gamma T_0$$

or $$\frac{T}{1-v/c} = \gamma T_0 .$$

Then

$$T = T_0 \frac{1-(v/c)}{\sqrt{1-(v^2/c^2)}} = T_0 \frac{1-(v/c)}{\sqrt{[1+(v/c)][1-(v/c)]}}$$

$$= \boxed{T_0 \sqrt{\frac{1-(v/c)}{1+(v/c)}}}$$

***P39.73** (a) The proper lifetime is measured in the ship's reference frame, and Earth-based observers measure a dilated lifetime of

$$\Delta t = \gamma \Delta t_p = \frac{\Delta t_p}{\sqrt{1-(v/c)^2}} = \frac{15.0 \text{ yr}}{\sqrt{1-(0.700)^2}} = \boxed{21.0 \text{ yr}}$$

(b) As measured by mission control, the distance to the ship is

$$d = v\Delta t = (0.700c)(21.0 \text{ yr}) = \left[(0.700)(1.00 \text{ ly/yr})\right](21.0 \text{ yr})$$

$$= \boxed{14.7 \text{ ly}}$$

(c) Looking out the rear window, the astronauts see Earth recede at a rate of $v = 0.700c$. The distance it has receded, as measured by the astronauts, when the batteries fail is

$$d = v(\Delta t_p) = (0.700c)(15.0 \text{ yr}) = \left[(0.700)(1.00 \text{ ly/yr})\right](15.0 \text{ yr})$$

$$= \boxed{10.5 \text{ ly}}$$

(d) Mission control gets signals for 21.0 yr while the battery is operating and then for 14.7 yr after the battery stops powering the transmitter, 14.7 ly away. The total time that signals are received is $21.0 \text{ yr} + 14.7 \text{ yr} = \boxed{35.7 \text{ yr}}$.

P39.74 (a) We let H represent K/mc^2. Then,

$$H + 1 = \frac{1}{\sqrt{1-u^2/c^2}}$$

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

so $\quad 1-u^2/c^2 = \dfrac{1}{H^2+2H+1}$

Solving,

$$\frac{u^2}{c^2} = 1 - \frac{1}{H^2+2H+1} = \frac{H^2+2H}{H^2+2H+1}$$

and $\boxed{u = c\left(\dfrac{H^2+2H}{H^2+2H+1}\right)^{1/2}}$

(b) $\boxed{u \text{ goes to 0 as } K \text{ goes to 0.}}$

(c) $\boxed{u \text{ approaches } c \text{ as } K \text{ increases without limit.}}$

(d) The acceleration is given by

$$a = \frac{du}{dt} = \frac{d}{dt}\left[c\left(\frac{H^2+2H}{H^2+2H+1}\right)^{1/2}\right]$$

$$a = c\frac{1}{2}\left(\frac{H^2+2H}{H^2+2H+1}\right)^{-1/2} \times\left(\frac{\left[H^2+2H+1\right][2H+2]-\left[H^2+2H\right][2H+2]}{\left[H^2+2H+1\right]^2}\right) \times\frac{d(K/mc^2)}{dt}$$

$$a = c\left(\frac{H^2+2H+1}{H^2+2H}\right)^{1/2}\left(\frac{H+1}{[H+1]^4}\right)\frac{P}{mc^2}$$

$$= \boxed{\frac{P}{mcH^{1/2}(H+2)^{1/2}(H+1)^2}}$$

where $P = \dfrac{dK}{dt}$.

(e) When H is small ($H \ll 1$), we have approximately

$$a = \frac{P}{mcH^{1/2}(2)^{1/2}(1)^2} = \frac{P}{mcH^{1/2}2^{1/2}} = \frac{P}{mc\left(\dfrac{K}{mc^2}\right)^{1/2}2^{1/2}}$$

$$= \frac{P}{(2mK)^{1/2}}$$

in agreement with the nonrelativistic case.

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

(f) When H is large the acceleration approaches

$$a = \frac{P}{mcH^{1/2}(H+2)^{1/2}(H+1)^2} \to \frac{P}{mcH^{1/2}(H)^{1/2}(H)^2} = \frac{P}{mcH^3}$$

$$= \frac{P}{mc\left(\frac{K}{mc^2}\right)^3} = \frac{m^2c^5P}{K^3}$$

(g) As energy is steadily imparted to the particle, the particle's acceleration decreases. It decreases steeply, proportionally to $1/K^3$ at high energy. In this way the particle's speed cannot reach or surpass a certain upper limit, which is the speed of light in vacuum.

***P39.75** (a) From Problem 71,

$$\frac{v}{c} = \frac{1}{\sqrt{1+2.25\times10^{-10}}}$$

and

$$\gamma = \frac{1}{\sqrt{1-\left(\frac{1}{\sqrt{1+2.25\times10^{-10}}}\right)^2}} = \frac{1}{\sqrt{1-\left(\frac{1}{1+2.25\times10^{-10}}\right)}}$$

$$= \boxed{6.67\times10^4}$$

(b) The astronaut's speed, from Problem 71, is

$$v = \frac{1}{\sqrt{1+2.25\times10^{-10}}}c$$

The time difference between the astronaut's trip and that of the beam of light is then

$$\Delta t = \frac{d}{v} - \frac{d}{c} = d\left(\frac{1}{v} - \frac{1}{c}\right) = \frac{d}{c}\left(\frac{c}{v} - 1\right) = \frac{d}{c}\left(\sqrt{1+x} - 1\right) \approx \frac{d}{c}\left(1 + \frac{x}{2} - 1\right)$$

$$= \frac{d}{c}\left(\frac{x}{2}\right)$$

Where $x = 2.25\times10^{-10}$. Substituting numerical values,

$$\Delta t = \frac{\left(2.00\times10^6 \text{ ly}\right)\left(9.46\times10^{15} \text{ m/ly}\right)}{3.00\times10^8 \text{ m/s}}\left(\frac{2.25\times10^{-10}}{2}\right)$$

$$= 7\ 095 \text{ s} = \boxed{1.96 \text{ h}}$$

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

P39.76 The energy of the first fragment is given by

$$E_1^2 = p_1^2c^2 + \left(m_1c^2\right)^2 = \left(1.75\text{ MeV}\right)^2 + \left(1.00\text{ MeV}\right)^2$$

$$E_1 = 2.02\text{ MeV}$$

For the second,

$$E_2^2 = \left(2.00\text{ MeV}\right)^2 + \left(1.50\text{ MeV}\right)^2$$

$$E_2 = 2.50\text{ MeV}$$

(a) Energy is conserved, so the unstable object had $E = E_1 + E_2 = 4.52\text{ MeV}$. Each component of momentum is conserved, so for the original object

$$p^2 = p_x^2 + p_y^2 = \left(\frac{1.75\text{ MeV}}{c}\right)^2 + \left(\frac{2.00\text{ MeV}}{c}\right)^2$$

Then, using Equation 39.27, we find the mass of the original object:

$$E^2 = p^2c^2 + \left(mc^2\right)^2$$

$$\left(4.52\text{ MeV}\right)^2 = \left[\left(1.75\text{ MeV}\right)^2 + \left(2.00\text{ MeV}\right)^2\right] + \left(mc^2\right)^2$$

$$\boxed{m = \frac{3.65\text{ MeV}}{c^2}}$$

(b) Now $E = \gamma mc^2$ gives

$$4.52\text{ MeV} = \frac{1}{\sqrt{1 - u^2/c^2}}\,3.65\text{ MeV}$$

$$1 - \frac{u^2}{c^2} = 0.654 \quad \text{which gives} \quad \boxed{u = 0.589c}$$

P39.77 The relativistic kinetic energy of such a proton is

$$K = \left(\gamma - 1\right)mc^2 = 10^{13}\text{ MeV}$$

Its rest energy is

$$mc^2 = \left(1.67 \times 10^{-27}\text{ kg}\right)\left(2.998 \times 10^8\text{ m/s}\right)^2 \times \left(\frac{1\text{ eV}}{1.60 \times 10^{-19}\text{ kg}\cdot\text{m}^2/\text{s}^2}\right) = 938\text{ MeV}$$

So $10^{13}\text{ MeV} = (\gamma - 1)(938\text{ MeV})$

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

and therefore, $\gamma = 1.07 \times 10^{10}$. The proton's speed in the galaxy's reference frame can be found from

$$\gamma = 1/\sqrt{1 - u^2/c^2} \qquad \text{so} \qquad 1 - u^2/c^2 = 8.80 \times 10^{-21}$$

and $$u = c\sqrt{1 - 8.80 \times 10^{-21}} \approx \left(1 - 4.40 \times 10^{-21}\right)c \approx 3.00 \times 10^8 \text{ m/s}$$

The proton's speed is nearly as large as the speed of light. In the galaxy frame, the traversal time is

$$\Delta t = x/u = 10^5 \text{ light years}/c = 10^5 \text{ years}$$

(a) This is dilated from the proper time measured in the proton's frame. The proper time interval is found from $\Delta t = \gamma \Delta t_p$:

$$\Delta t_p = \Delta t/\gamma = 10^5 \text{ yr}/1.07 \times 10^{10} = 9.38 \times 10^{-6} \text{ years} = 296 \text{ s}$$

Δt $\boxed{\sim 10^2 \text{ s or } 10^3 \text{ s}}$

(b) The proton sees the galaxy moving by at a speed nearly equal to c, passing in 296 s:

$$\Delta L_{\textit{proton frame}} = u\Delta t_p = \left(3.00 \times 10^8 \text{ m/s}\right)(296 \text{ s})$$
$$= 8.88 \times 10^7 \text{ km} \sim 10^8 \text{ km}$$

$$\Delta L_{\textit{proton frame}} = \left(8.88 \times 10^{10} \text{ m}\right)\left(\frac{1 \text{ ly}}{9.46 \times 10^{15} \text{ m}}\right)$$
$$= 9.39 \times 10^{-6} \text{ ly} \quad \boxed{\sim 10^{-5} \text{ ly}}$$

P39.78 Look at the situation from the instructors' viewpoint since they are at rest relative to the clock, and hence measure the proper time. The Earth moves with velocity $v = -0.280c$ relative to the instructors while the students move with a velocity $u' = -0.600c$ relative to Earth. Using the velocity addition equation, the velocity of the students relative to the instructors (and hence the clock) is:

$$u = \frac{v + u'}{1 + vu'/c^2} = \frac{\left(-0.280c\right) - \left(0.600c\right)}{1 + \left(-0.280c\right)\left(-0.600c\right)/c^2} = -0.753c$$

(students relative to clock)

(a) With a proper time interval of $\Delta t_p = 50.0 \text{ min}$, the time interval measured by the students is:

$$\Delta t = \gamma \Delta t_p \quad \text{with} \quad \gamma = \frac{1}{\sqrt{1 - \left(0.753c\right)^2/c^2}} = 1.52$$

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

Thus, the students measure the exam to last

$$T = 1.52(50.0 \text{ min}) = \boxed{76.0 \text{ minutes}}$$

(b) The duration of the exam as measured by observers on Earth is:

$$\Delta t = \gamma \Delta t_p \quad \text{with} \quad \gamma = \frac{1}{\sqrt{1 - \left(0.280c\right)^2 / c^2}} \quad \text{so}$$

$$T = 1.04(50.0 \text{ min}) = \boxed{52.1 \text{ minutes}}$$

P39.79 (a) The speed of light in water is $c/1.33$, so the electron's speed is $1.10c/1.333$. Then

$$\gamma = \frac{1}{\sqrt{1 - (1.10/1.333)^2}} = 1.770$$

and the total energy is

$$E = \gamma mc^2 = 1.770(0.511 \text{ MeV}) = \boxed{0.905 \text{ MeV}}$$

(b) The electron's kinetic energy is

$$K = E - mc^2 = 0.905 \text{ MeV} - 0.511 \text{ MeV} = \boxed{0.394 \text{ MeV}}$$

(c) The electron's momentum is found from

$$pc = \sqrt{E^2 - (mc^2)^2} = \sqrt{\gamma^2 - 1}\ mc^2$$
$$= \sqrt{\gamma^2 - 1}(0.511 \text{ MeV}) = 0.747 \text{ MeV}$$

and

$$p = \boxed{0.747 \frac{\text{MeV}}{c}} = \frac{0.747 \times 10^6 \left(1.602 \times 10^{-19} \text{ J}\right)}{3.00 \times 10^8 \text{ m/s}}$$
$$= \boxed{3.99 \times 10^{-22} \text{ kg} \cdot \text{m/s}}$$

(d) From Figure 17.11, the angle between the particle (source of waves) and the shock wave is

$$\sin\theta = v/v_S$$

where v is the wave speed, which is the speed of light in water, and v_S is the source speed. Then

$$\sin\theta = v/v_S = 1/1.10 \quad \rightarrow \quad \theta = \boxed{65.4^\circ}$$

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

P39.80 (a) From Equation 39.18, the speed of light in the laboratory frame is

$$u = \frac{v + \dfrac{c}{n}}{1 + \dfrac{v\left(c/n\right)}{c^2}} = \frac{c\ (1 + nv/c)}{n\ (1 + v/nc)}$$

(b) When v is much less than c we have

$$u = \frac{c}{n}\left(1 + \frac{nv}{c}\right)\left(1 + \frac{v}{nc}\right)^{-1} \approx \frac{c}{n}\left(1 + \frac{nv}{c}\right)\left(1 - \frac{v}{nc}\right)$$

$$\approx \frac{c}{n}\left(1 + \frac{nv}{c} - \frac{v}{nc}\right) = \frac{c}{n} + v - \frac{v}{n^2}$$

(c) If light travels at speed c/n in the water, and the water travels at speed v, then the Galilean velocity transformation Equation 4.20 would indeed give $c/n + v$ for the speed of light in the moving water. The third term $-v/n^2$ does represent a relativistic effect that was observed decades before the Michelson-Morley experiment. It is a piece of twentieth-century physics that dropped into the nineteenth century. We could say that light is intrinsically relativistic.

(d) To take the limit as v approaches c we must go back to $u = \dfrac{c\ (1 + nv/c)}{n\ (1 + v/nc)}$. As $v \to c$,

$$u \to \frac{c\ (1 + nc/c)}{n\ (1 + c/nc)} = \frac{c(1+n)}{n+1} = \boxed{c}$$

P39.81 (a) Assuming the Sun-mass system is isolated, the energy (work) required to remove a mass m from the Sun's surface to infinity is equal to the change in potential energy of the system. If the work equals the rest energy mc^2, then

$$W = \Delta E = \Delta K + \Delta U = 0 + \left(U_f - U_i\right)$$

$$mc^2 = 0 - \left(-\frac{GM_s m}{R_g}\right)$$

$$mc^2 = \frac{GM_s m}{R_g} \quad \rightarrow \quad R_g = \frac{GM_s}{c^2}$$

(b) $$R_g = \frac{GM_s}{c^2} = \frac{\left(6.67 \times 10^{-11}\ \text{N.m}^2/\text{kg}^2\right)\left(1.99 \times 10^{30}\ \text{kg}\right)}{\left(3.00 \times 10^8\ \text{m/s}\right)^2}$$

$$R_g = 1.47 \times 10^3\ \text{m} = \boxed{1.47\ \text{km}}$$

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

P39.82 We find the speed of the electrons after accelerating through a potential difference ΔV from Equation 39.23:

$$K = e\Delta V = (\gamma - 1)mc^2 = \left(\frac{1}{\sqrt{1 - (u/c)^2}} - 1 \right) mc^2$$

then

$$\frac{1}{\sqrt{1 - (u/c)^2}} = \frac{e\Delta V}{mc^2} + 1 = \frac{e\Delta V + mc^2}{mc^2}$$

or

$$1 - (u/c)^2 = \left(\frac{mc^2}{e\Delta V + mc^2} \right)^2$$

Solving,

$$\frac{u}{c} = \sqrt{1 - \left(\frac{\mathrm{m}}{\mathrm{e\Delta V/c^2 + m}} \right)^2}$$

Substituting numerical values and suppressing units,

$$\frac{u}{c} = \sqrt{1 - \left[\frac{(9.11 \times 10^{-31}\ \mathrm{kg})}{\frac{(1.60 \times 10^{-19}\ \mathrm{C})(8.40 \times 10^{4}\ \mathrm{V})}{(3.00 \times 10^{8}\ \mathrm{m/s})^2} + 9.11 \times 10^{-31}\ \mathrm{kg}} \right]^2}$$

$$u = 0.512c$$

Because this speed is more than half the speed of light, there is no way to double its speed, regardless of the increased accelerating voltage. If the accelerating voltage is quadrupled to 336 kV, the speed of the electrons rises to $u = 0.798c$.

P39.83 (a) Take the spaceship as the primed frame, moving toward the right at $v = +0.600c$. Then $u_x' = +0.800c$, and

$$u_x = \frac{u_x' + v}{1 + (u_x' v)/c^2} = \frac{0.800c + 0.600c}{1 + (0.800)(0.600)} = \boxed{0.946c}$$

(b) $L = \dfrac{L_p}{\gamma}$: $\quad L = (0.200\ \mathrm{ly})\sqrt{1 - (0.600)^2} = \boxed{0.160\ \mathrm{ly}}$

(c) The aliens observe the 0.160-ly interval decreasing because the probe reduces it from one end at $0.800c$ and the Earth reduces it at the other end at $0.600c$.

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

Thus, $$\text{time} = \frac{0.160 \text{ ly}}{0.800c + 0.600c} = \boxed{0.114 \text{ yr}}$$

(d) In Earth's reference frame, the kinetic energy of the landing craft is

$$K = \left(\frac{1}{\sqrt{1 - u^2/c^2}} - 1 \right) mc^2$$

$$K = \left(\frac{1}{\sqrt{1 - (0.946)^2}} - 1 \right) \left(4.00 \times 10^5 \text{ kg}\right)\left(3.00 \times 10^8 \text{ m/s}\right)^2$$

$$= \boxed{7.50 \times 10^{22} \text{ J}}$$

P39.84 (a) Take $m = 1.00$ kg. The classical kinetic energy is

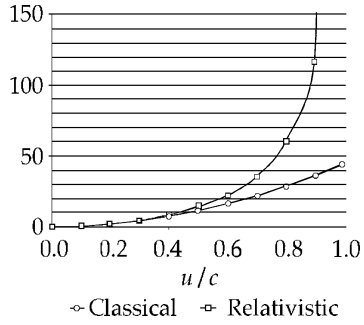
$$K_c = \frac{1}{2}mu^2 = \frac{1}{2}mc^2\left(\frac{u}{c}\right)^2 = \left(4.50 \times 10^{16} \text{ J}\right)\left(\frac{u}{c}\right)^2$$

and the actual kinetic energy is

$$K_r = \left(\frac{1}{\sqrt{1 - (u/c)^2}} - 1 \right) mc^2 = \left(9.00 \times 10^{16} \text{ J}\right)\left(\frac{1}{\sqrt{1 - (u/c)^2}} - 1 \right)$$

Using these expressions, we generate the graph in ANS. GRAPH P39.84.

$\frac{u}{c}$	K_c (J)	K_r (J)
0.000	0.000	0.000
0.100	0.045×10^{16}	$0.045\,3 \times 10^{16}$
0.200	0.180×10^{16}	0.186×10^{16}
0.300	0.405×10^{16}	0.435×10^{16}
0.400	0.720×10^{16}	0.820×10^{16}
0.500	1.13×10^{16}	1.39×10^{16}
0.600	1.62×10^{16}	2.25×10^{16}
0.700	2.21×10^{16}	3.60×10^{16}
0.800	2.88×10^{16}	6.00×10^{16}
0.900	3.65×10^{16}	11.6×10^{16}
0.990	4.41×10^{16}	54.8×10^{16}



ANS. GRAPH P39.84

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

(b) $K_c = 0.990K_r$, when $\frac{1}{2}\left(\frac{u}{c}\right)^2 = 0.990\left[\frac{1}{\sqrt{1-(u/c)^2}} - 1\right]$, yielding

$$u = \boxed{0.115c}$$

(c) Similarly, $K_c = 0.950K_r$ when $u = \boxed{0.257c}$.

(d) $K_c = 0.500K_r$ when $u = \boxed{0.786c}$.

P39.85 Both observers measure the speed of light to be c.

(a) Call the total travel time Δt_S. An observer at rest relative to the mirror sees the light travel a distance $d_1 = d$ from the spacecraft to the mirror, but a distance $d_2 = d - v\Delta t_S$ from the mirror back to the spacecraft because the spacecraft has traveled the distance $v\Delta t_S$ forward. Therefore, the total distance traveled by the light is

$$D = d_1 + d_2$$
$$d + (d - v\Delta t_S) = c\Delta t_S$$
$$\Delta t_S = \frac{2d}{c+v} = \frac{2d}{c+0.650c} = \frac{2(5.66\times10^{10}\ \text{m})}{1.650(3.00\times10^{8}\ \text{m/s})} = \boxed{229\ \text{s}}$$

(b) The observer in the spacecraft measures a length-contracted initial distance to the mirror of

$$L = d\sqrt{1-\frac{v^2}{c^2}}$$

and the mirror moving toward the ship at speed v. Consider the motion of the light toward the mirror in time interval Δt_1: light travels toward the mirror at speed c while the mirror travels toward the spacecraft at speed v; together, they travel the distance L:

$$c\Delta t_1 + v\Delta t_1 = L$$
$$\Delta t_1 = \frac{L}{c+v}$$

When light strikes the mirror, it is a distance $L' = L - v\Delta t_1$ from the spacecraft. The light must travel back through this same distance to return to the spacecraft:

$$c\Delta t_2 = L - v\Delta t_1 \quad \rightarrow \quad \Delta t_2 = \frac{L}{c} - \frac{v}{c}\Delta t_1$$

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

The total travel time is

$$\Delta t_1 + \Delta t_2 = \frac{L}{c+v} + \frac{L}{c} - \frac{v}{c}\Delta t_1 = \frac{L}{c+v} + \frac{L}{c} - \frac{v}{c}\left(\frac{L}{c+v}\right)$$

$$= \frac{Lc + L(c+v) - Lv}{c(c+v)} = \frac{2L\cancel{c}}{\cancel{c}(c+v)} = \frac{2}{(c+v)}d\sqrt{1-\frac{v^2}{c^2}}$$

$$= \frac{2}{(c+v)}d\frac{\sqrt{c^2-v^2}}{c}$$

$$\Delta t_1 + \Delta t_2 = \frac{2d}{c}\sqrt{\frac{c-v}{c+v}} = \frac{2d}{c}\sqrt{\frac{c-0.650c}{c+0.650c}}$$

$$= \frac{2\left(5.66\times10^{10}\text{ m}\right)}{\left(3.00\times10^{8}\text{ m/s}\right)}\sqrt{\frac{0.350}{1.650}}$$

$$= \boxed{174\text{ s}}$$

P39.86 Both observers measure the speed of light to be c.

(a) Call the total travel time Δt_S. An observer at rest relative to the mirror sees the light travel a distance $d_1 = d$ from the spacecraft to the mirror, but a distance $d_2 = d - v\Delta t_S$ from the mirror back to the spacecraft because the spacecraft has traveled the distance $v\Delta t_S$ forward. Therefore, the total distance traveled by the light is

$$D = d_1 + d_2$$

$$d + \left(d - v\Delta t_S\right) = c\Delta t_S$$

$$\Delta t_S = \boxed{\frac{2d}{c+v}}$$

(b) The observer in the spacecraft measures a length-contracted initial distance to the mirror of

$$L = d\sqrt{1-\frac{v^2}{c^2}}$$

and the mirror moving toward the ship at speed v. Consider the motion of the light toward the mirror in time interval Δt_1: light travels toward the mirror at speed c while the mirror travels toward the spacecraft at speed v; together, they travel the distance L:

$$c\Delta t_1 + v\Delta t_1 = L$$

$$\Delta t_1 = \frac{L}{c+v}$$

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

When light strikes the mirror, it is a distance $L' = L - v\Delta t_1$ from the spacecraft. The light must travel back through this same distance to return to the spacecraft:

$$c\Delta t_2 = L - v\Delta t_1 \quad \rightarrow \quad \Delta t_2 = \frac{L}{c} - \frac{v}{c}\Delta t_1$$

The total travel time is

$$\begin{aligned}\Delta t_1 + \Delta t_2 &= \frac{L}{c+v} + \frac{L}{c} - \frac{v}{c}\Delta t_1 \\ &= \frac{L}{c+v} + \frac{L}{c} - \frac{v}{c}\left(\frac{L}{c+v}\right) \\ &= \frac{Lc + L(c+v) - Lv}{c(c+v)} = \frac{2L\cancel{c}}{\cancel{c}(c+v)} = \frac{2}{(c+v)}d\sqrt{1-\frac{v^2}{c^2}} \\ &= \frac{2}{(c+v)}d\frac{\sqrt{c^2-v^2}}{c}\end{aligned}$$

$$\Delta t_1 + \Delta t_2 = \boxed{\frac{2d}{c}\sqrt{\frac{c-v}{c+v}}}$$

P39.87 Since the total momentum is zero before decay, it is necessary that after the decay

$$p_{\text{nucleus}} = p_{\text{photon}} = \frac{E_\gamma}{c} = \frac{14.0 \text{ keV}}{c}$$

Also, for the recoiling nucleus, $E^2 = p^2c^2 + \left(mc^2\right)^2$ with

$$Mc^2 = 8.60 \times 10^{-9} \text{ J} = 5.38 \times 10^{10} \text{ eV} = 5.38 \times 10^7 \text{ keV}$$

Thus, $$\left(Mc^2 + K\right)^2 = \left(14.0 \text{ keV}\right)^2 + \left(Mc^2\right)^2$$

or

$$\left(1 + \frac{K}{Mc^2}\right)^2 = \left(\frac{14.0 \text{ keV}}{Mc^2}\right)^2 + 1$$

Because the term $\left(\frac{14.0 \text{ keV}}{Mc^2}\right)^2 \ll 1$, evaluating $\left(\frac{14.0 \text{ keV}}{Mc^2}\right)^2 + 1$ on a calculator gives 1.

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

We need to expand $\left(\frac{14.0\text{ keV}}{Mc^2}\right)^2 + 1$ using the Binomial Theorem:

$$1+\frac{K}{Mc^2}=\sqrt{1+\left(\frac{14.0\text{ keV}}{Mc^2}\right)^2}\approx 1+\frac{1}{2}\left(\frac{14.0\text{ keV}}{Mc^2}\right)^2$$

$$K\approx\frac{(14.0\text{ keV})^2}{2Mc^2}=\frac{\left(14.0\times10^3\text{ eV}\right)^2}{2\left(53.8\times10^9\text{ eV}\right)}=\boxed{1.82\times10^{-3}\text{ eV}}$$

Challenge Problems

P39.88 (a) At any speed, the momentum of the particle is given by

$$p=\gamma mu=\frac{mu}{\sqrt{1-(u/c)^2}}$$

With Newton's law expressed as $F=qE=\frac{dp}{dt}$, we have

$$qE=\frac{d}{dt}\left[mu\left(1-\frac{u^2}{c^2}\right)^{-1/2}\right]$$

$$qE=m\left(1-\frac{u^2}{c^2}\right)^{-1/2}\frac{du}{dt}+\frac{1}{2}mu\left(1-\frac{u^2}{c^2}\right)^{-3/2}\left(\frac{2u}{c^2}\right)\frac{du}{dt}$$

so $$\frac{qE}{m}=\frac{du}{dt}\left[\frac{1-u^2/c^2+u^2/c^2}{\left(1-u^2/c^2\right)^{3/2}}\right]$$

and $$\boxed{a=\frac{du}{dt}=\frac{qE}{m}\left(1-\frac{u^2}{c^2}\right)^{3/2}}$$

(b) For u small compared to c, the relativistic expression reduces to the classical $a=\frac{qE}{m}$. As u approaches c, the acceleration approaches zero, so that the object can never reach the speed of light.

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

(c) We can use the result of (a) to find the velocity u at time t:

$$a = \frac{du}{dt} = \frac{qE}{m}\left(1 - \frac{u^2}{c^2}\right)^{3/2} \quad \rightarrow \quad \int_0^u \frac{du}{\left(1 - u^2/c^2\right)^{3/2}} = \int_0^t \frac{qE}{m} dt$$

$$\frac{u}{\left(1 - u^2/c^2\right)^{1/2}} = \frac{qEt}{m}$$

$$u^2 = \left(\frac{qEt}{m}\right)^2 \left(1 - \frac{u^2}{c^2}\right)$$

$$\boxed{u = \frac{qEct}{\sqrt{m^2c^2 + q^2E^2t^2}}}$$

Now, we can use this result to find position x at time t:

$$\frac{dx}{dt} = u = \frac{qEct}{\sqrt{m^2c^2 + q^2E^2t^2}}$$

$$x = \int_0^t u dt = qEc \int_0^t \frac{tdt}{\sqrt{m^2c^2 + q^2E^2t^2}} = \frac{c}{qE}\sqrt{m^2c^2 + q^2E^2t^2}\Big|_0^t$$

$$\boxed{x = \frac{c}{qE}\left(\sqrt{m^2c^2 + q^2E^2t^2} - mc\right)}$$

P39.89 (a) Take the two colliding protons as the system

$$E_1 = K + mc^2 \qquad E_2 = mc^2$$

$$E_1^2 = p_1^2c^2 + m^2c^4 \qquad p_2 = 0$$

In the final state,

$$E_f = K_f + Mc^2 = p_f^2c^2 + M^2c^4$$

By energy conservation, $E_1 + E_2 = E_f$, so

$$E_1^2 + 2E_1E_2 + E_2^2 = E_f^2$$

$$p_1^2c^2 + m^2c^4 + 2\left(K + mc^2\right)mc^2 + m^2c^4$$
$$= p_f^2c^2 + M^2c^4$$

By conservation of momentum, $p_1 = p_f$, so

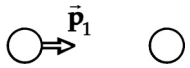
$$\cancel{p_1^2c^2} + m^2c^4 + 2\left(K + mc^2\right)mc^2 + m^2c^4$$
$$= \cancel{p_f^2c^2} + M^2c^4$$

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

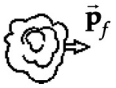
and we have then

$$M^2c^4 = 2Kmc^2 + 4m^2c^4 = \frac{4Km^2c^4}{2mc^2} + 4m^2c^4$$

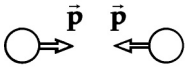
$$Mc^2 = 2mc^2\sqrt{1+\frac{K}{2mc^2}}$$



initial



final



initial (beams)



final (beams)

ANS. FIG. P39.89

(b) By contrast, for colliding beams we have, in the original state,

$$E_1 = K + mc^2 \qquad E_2 = K + mc^2$$

In the final state,

$$E_f = Mc^2$$

$E_1 + E_2 = E_f$:

$$K + mc^2 + K + mc^2 = Mc^2$$

$$Mc^2 = 2mc^2\left(1+\frac{K}{2mc^2}\right)$$

P39.90 We choose to write down the answer to part (b) first.

(b) Consider a hermit who lives on an asteroid halfway between the Sun and Tau Ceti, stationary with respect to both. Just as our spaceship is passing him, he also sees the blast waves from both explosions. Judging both stars to be stationary, this observer concludes that the two stars blew up simultaneously.

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

(a) We in the spaceship moving past the hermit do not calculate the explosions to be simultaneous. We measure the distance we have traveled from the Sun as

$$L = L_p\sqrt{1-\left(\frac{v}{c}\right)^2} = (6.00\text{ ly})\sqrt{1-(0.800)^2} = 3.60\text{ ly}$$

We see the Sun flying away from us at $0.800c$ while the light from the Sun approaches at $1.00c$. Thus, the gap between the Sun and its blast wave has opened at $1.80c$, and the time we calculate to have elapsed since the Sun exploded is

$$\frac{3.60\text{ ly}}{1.80c} = 2.00\text{ yr}$$

We see Tau Ceti as moving toward us at $0.800c$, while its light approaches at $1.00c$, only $0.200c$ faster. We measure the gap between that star and its blast wave as 3.60 ly and growing at $0.200c$. We calculate that it must have been opening for

$$\frac{3.60\text{ ly}}{0.200c} = 18.0\text{ yr}$$

and conclude that $\boxed{\text{Tau Ceti exploded 16.0 years before the Sun}}$.

P39.91 (a) Since Dina is in the same reference frame, S′, as Owen, she measures the ball to have the same speed Owen observes, namely

$$|u_x'| = \boxed{0.800c}$$

(b) Within the frame S′, the ball travels 1.80×10^{12} m at a speed of $0.800c$, so

$$\Delta t' = \frac{L_p}{|u_x'|} = \frac{1.80\times10^{12}\text{ m}}{0.800\left(3.00\times10^8\text{ m/s}\right)} = \boxed{7.50\times10^3\text{ s}}$$

(c) In the S frame, the distance between Dina and Owen is a proper length; therefore,

$$L = L_p\sqrt{1-\frac{v^2}{c^2}} = \left(1.80\times10^{12}\text{ m}\right)\sqrt{1-\frac{\left(0.600c\right)^2}{c^2}} = \boxed{1.44\times10^{12}\text{ m}}$$

Since $v = 0.600c$ and $u_x' = -0.800c$, the velocity Ed measures for the ball is

$$u_x = \frac{u_x' + v}{1 + u_x'v/c^2} = \frac{\left(-0.800c\right)+\left(0.600c\right)}{1+\left(-0.800\right)\left(0.600\right)} = \boxed{-0.385c}$$

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

(d) Ed measures the ball and Dina to be initially separated by 1.44×10^{12} m. Dina's motion at $0.600c$ and the ball's motion at $0.385c$ cover this distance from both ends. The gap closes at the rate $0.600c + 0.385c = 0.985c$, so the ball and catcher meet after a time

$$\Delta t = \frac{1.44 \times 10^{12} \text{ m}}{0.985\left(3.00 \times 10^{8} \text{ m/s}\right)} = \boxed{4.88 \times 10^{3} \text{ s}}$$

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

ANSWERS TO EVEN-NUMBERED PROBLEMS

P39.2 (a–b) See P39.2 for full explanation.

P39.4 See P39.4 for full explanation.

P39.6 (a) 0.436 m; (b) less than 0.436 m

P39.8 (a) $2.18\ \mu s$; (b) 649 m

P39.10 65.0 beats/min; (b) 10.5 beats/min

P39.12 (a) $L_p = 20.0$ m; (b) $L = 19.0$ m; (c) $0.312c$

P39.14 $0.140c$

P39.16 (a) 1.3×10^{-7} s; (b) 38 m; (c) 7.6 m

P39.18 42.1 g/cm^3

P39.20 $$v = \frac{cL_p}{\sqrt{c^2\Delta t^2 + L_p^2}}$$

P39.22 (a) $39.2\ \mu s$; (b) accurate to one digit

P39.24 (a) 5.45 yr; (b) Goslo

P39.26 1.13×10^4 Hz

P39.28 (a) $v = 0.943c$; (b) 2.55×10^3 m

P39.30 (a) $L = L_0\left[1-\left(\frac{v}{c}\right)^2\cos^2\theta_0\right]^{1/2}$; (b) $\gamma\tan\theta_0$

P39.32 $0.960c$

P39.34 0.893c, 16.8° above the x' axis

P39.36 (a) $2.73\times10^{-24}\ \text{kg}\cdot\text{m/s}$; (b) $1.58\times10^{-22}\ \text{kg}\cdot\text{m/s}$;
(c) $5.64\times10^{-22}\ \text{kg}\cdot\text{m/s}$

P39.38 $$u = \frac{c}{\sqrt{(m^2c^2/p^2)+1}}$$

P39.40 (a) \$800; (b) $\$2.12 \times 10^9$

P39.42 (a) $0.141c$; (b) $0.436c$

P39.44 (a) 0.582 MeV; (b) 2.45 MeV

P39.46 (a) $0.999997c$; (b) 3.74×10^5 MeV

P39.48 (a) 4.38×10^{11} J; (b) 4.38×10^{11}; (c) See P39.48(c) for full explanation.

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

P39.50 See P39.50 for full explanation.

P39.52 (a) 3.91×10^4; (b) $0.9999999997c$; (c) 7.67 cm

P39.54 0.842 kg

P39.56 1.20 MeV

P39.58 See P39.58 for full explanation.

P39.60 larger; $\sim 10^{-9}$ J

P39.62 (a) isolated; (b) isolated system: conservation of energy and isolated system: conservation of momentum; (c) 6.22 and 2.01;
(d) $3.09m_1 + m_2 = 1.66 \times 10^{-27}$ kg ; (e) $m_2 = 3.52m_1$;
(f) $m_1 = 2.51 \times 10^{-28}$ kg and $m_2 = 8.84 \times 10^{-28}$ kg

P39.64 (a) $M = \dfrac{2m\sqrt{4 - u^2/c^2}}{3\sqrt{1 - u^2/c^2}}$; (b) $\dfrac{4m}{3}$; (c) The answer to part (b) is in agreement with the classical result, which is the arithmetic sum of the masses of the two colliding particles.

P39.66 (a) $0.023\,6c$; (b) $6.18 \times 10^{-4}c$

P39.68 When Speedo arrives back on Earth, 118 years have passed, and Goslo would be 158 years old. That is impossible at the present time.

P39.70 (a) $0.467c$; (b) 2.75×10^3 kg

P39.72 See P39.72 for full explanation.

P39.74 (a) $u = c\left(\dfrac{H^2 + 2H}{H^2 + 2H + 1}\right)^{1/2}$; (b) u goes to 0 as K goes to 0; (c) u approaches c as K increases without limit; (d) $\dfrac{P}{mcH^{1/2}(H+2)^{1/2}(H+1)^2}$;
(e) See P39.74(e) for full explanation; (f) See P39.74(f) for full explanation; (g) As energy is steadily imparted to particle, the particle's acceleration decreases. It decreases steeply, proportionally to $1/K^3$ at high energy. In this way the particle's speed cannot reach or surpass a certain upper limit, which is the speed of light in vacuum.

P39.76 (a) $m = \dfrac{3.65 \text{ MeV}}{c^2}$; (b) $v = 0.589c$

P39.78 (a) 76.0 minutes; (b) 52.1 minutes

P39.80 (a–c) See P39.80 for full explanation; (d) c

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.

P39.82 Because the speed of the electrons after accelerating through a potential difference ΔV is more than half the speed of light, there is no way to double its speed, regardless of the increased accelerating voltage.

P39.84 (a) See ANS. GRAPH P39.84; (b) 0.115*c*; (c) 0.257*c*; (d) 0.786*c*

P39.86 (a) $\frac{2d}{c+v}$; (b) $\frac{2d}{c}\sqrt{\frac{c-v}{c+v}}$

P39.88 (a) $a = \frac{du}{dt} = \frac{qE}{m}\left(1 - \frac{u^2}{c^2}\right)^{3/2}$; (b) For u small compared to c, the relativistic expression reduces to the classical $a = \frac{qE}{m}$. As u approaches c, the acceleration approaches zero, so that the object can never reach the speed of light; (c) $u = \frac{qEct}{\sqrt{m^2c^2 + q^2E^2t^2}}$ and $x = \frac{c}{qE}\left(\sqrt{m^2c^2 + q^2E^2t^2} - mc\right)$

P39.90 (a) Tau Ceti exploded 16.0 years before the Sun; (b) The two stars blew up simultaneously.

© 2014 Cengage Learning. All Rights Reserved. May not be scanned, copied or duplicated, or posted to a publicly accessible website, in whole or in part.